



ANSI/TIA-102.BAAD-B-2015
APPROVED: JULY 28, 2015

TIA STANDARD

Project 25 Conventional Procedures

**TIA-102.BAAD-B
(Revision of TIA-102.BAAD-A)**

August 2015

**TELECOMMUNICATIONS
INDUSTRY ASSOCIATION**

tiaonline.org

NOTICE

TIA Engineering Standards and Publications are designed to serve the public interest through eliminating misunderstandings between manufacturers and purchasers, facilitating interchangeability and improvement of products, and assisting the purchaser in selecting and obtaining with minimum delay the proper product for their particular need. The existence of such Standards and Publications shall not in any respect preclude any member or non-member of TIA from manufacturing or selling products not conforming to such Standards and Publications. Neither shall the existence of such Standards and Publications preclude their voluntary use by Non-TIA members, either domestically or internationally.

Standards and Publications are adopted by TIA in accordance with the American National Standards Institute (ANSI) patent policy. By such action, TIA does not assume any liability to any patent owner, nor does it assume any obligation whatever to parties adopting the Standard or Publication.

This Standard does not purport to address all safety problems associated with its use or all applicable regulatory requirements. It is the responsibility of the user of this Standard to establish appropriate safety and health practices and to determine the applicability of regulatory limitations before its use.

Any use of trademarks in this document are for information purposes and do not constitute an endorsement by TIA or this committee of the products or services of the company.

(From Project No. ANSI/TIA-PN-102.BAAD-B, formulated under the cognizance of the TIA TR-8 Mobile and Personal Private Radio Standards, TR-8.15 Subcommittee on Common Air Interface).

Published by

©TELECOMMUNICATIONS INDUSTRY ASSOCIATION
Technology and Standards Department
1320 N. Courthouse Road, Suite 200
Arlington, VA 22201 U.S.A.

**PRICE: Please refer to current Catalog of
TIA TELECOMMUNICATIONS INDUSTRY ASSOCIATION STANDARDS
AND ENGINEERING PUBLICATIONS
or call IHS, USA and Canada
(1-877-413-5187) International (303-397-2896)
or search online at <http://www.tiaonline.org/standards/catalog/>**

All rights reserved
Printed in U.S.A.

NOTICE OF COPYRIGHT

This document is copyrighted by the TIA.

Reproduction of these documents either in hard copy or soft copy (including posting on the web) is prohibited without copyright permission. For copyright permission to reproduce portions of this document, please contact the TIA Standards Department or go to the TIA website (www.tiaonline.org) for details on how to request permission. Details are located at:

<http://www.tiaonline.org/standards/catalog/info.cfm#copyright>

or

Telecommunications Industry Association
Technology & Standards Department
1320 N. Courthouse Road, Suite 200
Arlington, VA 22201 USA
+1.703.907.7700

Organizations may obtain permission to reproduce a limited number of copies by entering into a license agreement. For information, contact

IHS
15 Inverness Way East
Englewood, CO 80112-5704
or call
USA and Canada (1.800.525.7052)
International (303.790.0600)

Limited Use Only

NOTICE OF DISCLAIMER AND LIMITATION OF LIABILITY

The document to which this Notice is affixed (the "Document") has been prepared by one or more Engineering Committees or Formulating Groups of the Telecommunications Industry Association ("TIA"). TIA is not the author of the Document contents, but publishes and claims copyright to the Document pursuant to licenses and permission granted by the authors of the contents.

TIA Engineering Committees and Formulating Groups are expected to conduct their affairs in accordance with the TIA Procedures for American National Standards and TIA Engineering Committee Operating Procedures, the current and predecessor versions of which are available at <http://www.tiaonline.org/standards/ec-procedures>/TIA's function is to administer the process, but not the content, of document preparation in accordance with the Manual and, when appropriate, the policies and procedures of the American National Standards Institute ("ANSI"). TIA does not evaluate, test, verify or investigate the information, accuracy, soundness, or credibility of the contents of the Document. In publishing the Document, TIA disclaims any undertaking to perform any duty owed to or for anyone.

If the Document is identified or marked as a project number (PN) document, or as a standards proposal (SP) document, persons or parties reading or in any way interested in the Document are cautioned that: (a) the Document is a proposal; (b) there is no assurance that the Document will be approved by any Committee of TIA or any other body in its present or any other form; (c) the Document may be amended, modified or changed in the standards development or any editing process.

The use or practice of contents of this Document may involve the use of intellectual property rights ("IPR"), including pending or issued patents, or copyrights, owned by one or more parties. TIA makes no search or investigation for IPR. When IPR consisting of patents and published pending patent applications are claimed and called to TIA's attention, a statement from the holder thereof is requested, all in accordance with the Manual. TIA takes no position with reference to, and disclaims any obligation to investigate or inquire into, the scope or validity of any claims of IPR. TIA will neither be a party to discussions of any licensing terms or conditions, which are instead left to the parties involved, nor will TIA opine or judge whether proposed licensing terms or conditions are reasonable or non-discriminatory. TIA does not warrant or represent that procedures or practices suggested or provided in the Manual have been complied with as respects the Document or its contents.

If the Document contains one or more Normative References to a document published by another organization ("other SSO") engaged in the formulation, development or publication of standards (whether designated as a standard, specification, recommendation or otherwise), whether such reference consists of mandatory, alternate or optional elements (as defined in the TIA Procedures for American National Standards) then (i) TIA disclaims any duty or obligation to search or investigate the records of any other SSO for IPR or letters of assurance relating to any such Normative Reference; (ii) TIA's policy of encouragement of voluntary disclosure (see TIA Procedures for American National Standards Annex C.1.2.3) of Essential Patent(s) and published pending patent applications shall apply; and (iii) Information as to claims of IPR in the records or publications of the other SSO shall not constitute identification to TIA of a claim of Essential Patent(s) or published pending patent applications.

TIA does not enforce or monitor compliance with the contents of the Document. TIA does not certify, inspect, test or otherwise investigate products, designs or services or any claims of compliance with the contents of the Document.

ALL WARRANTIES, EXPRESS OR IMPLIED, ARE DISCLAIMED, INCLUDING WITHOUT LIMITATION, ANY AND ALL WARRANTIES CONCERNING THE ACCURACY OF THE CONTENTS, ITS FITNESS OR APPROPRIATENESS FOR A PARTICULAR PURPOSE OR USE, ITS MERCHANTABILITY AND ITS NONINFRINGEMENT OF ANY THIRD PARTY'S INTELLECTUAL PROPERTY RIGHTS. TIA EXPRESSLY DISCLAIMS ANY AND ALL RESPONSIBILITIES FOR THE ACCURACY OF THE CONTENTS AND MAKES NO REPRESENTATIONS OR WARRANTIES REGARDING THE CONTENT'S COMPLIANCE WITH ANY APPLICABLE STATUTE, RULE OR REGULATION, OR THE SAFETY OR HEALTH EFFECTS OF THE CONTENTS OR ANY PRODUCT OR SERVICE REFERRED TO IN THE DOCUMENT OR PRODUCED OR RENDERED TO COMPLY WITH THE CONTENTS.

TIA SHALL NOT BE LIABLE FOR ANY AND ALL DAMAGES, DIRECT OR INDIRECT, ARISING FROM OR RELATING TO ANY USE OF THE CONTENTS CONTAINED HEREIN, INCLUDING WITHOUT LIMITATION ANY AND ALL INDIRECT, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES (INCLUDING DAMAGES FOR LOSS OF BUSINESS, LOSS OF PROFITS, LITIGATION, OR THE LIKE), WHETHER BASED UPON BREACH OF CONTRACT, BREACH OF WARRANTY, TORT (INCLUDING NEGLIGENCE), PRODUCT LIABILITY OR OTHERWISE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. THE FOREGOING NEGATION OF DAMAGES IS A FUNDAMENTAL ELEMENT OF THE USE OF THE CONTENTS HEREOF, AND THESE CONTENTS WOULD NOT BE PUBLISHED BY TIA WITHOUT SUCH LIMITATIONS.

This copy is provided to a TIA Member. It is not to be shared or posted to a web server accessible to others.

To purchase a copy of this standard go to <https://store.accuristech.com/tia> or contact TIA at standards@tiaonline.org.

Limited Use Only

Foreword

(This foreword is not part of this document.)

This document has been created in response to a request by the Project 25 Steering Committee as provided for in a Memorandum of Understanding (MOU) dated April 1992 and amended December 1993. The MOU states that the Project 25 Steering Committee shall devise a Common System Standard for digital public safety communications (the Standard) commonly referred to as Project 25 or P25, and that TIA, as agreed to by the membership of the appropriate TIA Engineering Committee, will provide technical assistance in the development of documentation for the Standard. If the abbreviation "P25" or the wording "Project 25" appears on the cover sheet of this document when published that indicates the Project 25 Steering Committee has adopted this document as part of the Standard. The appearance of the abbreviation "P25" or the wording "Project 25" on the cover sheet of this document or in the title of any document referenced herein should not be interpreted as limiting the applicability of the information contained in any document to "P25" or "Project 25" implementations exclusively.

This document was developed by the TR-8.15 Common Air Interface Subcommittee and provides a specification of the Conventional Procedures for TIA-102 Conventional LMR systems.

The 3rd edition (Revision B) of this document revised the Conventional Procedures specific to TIA-102 Packet Data to ensure the procedures are complete in order to facilitate interoperability and the development of packet data test documents. The revision also removed the Packet Data Logical Link Control Procedures which are now published in TIA-102.BAED and incorporated errata updates. This document cancels and replaces TIA-102.BAAD-A.

There are four annexes in this document. Annexes A, B, C, and D are informative and are not considered part of this specification.

TIA does not evaluate, test, verify or investigate the information, accuracy, soundness, or credibility of the contents of the Document. In Publishing the Document, TIA disclaims any undertaking to perform any duty owed to or for anyone. Some aspects of the specifications contained in this document may not have been fully operationally tested; however, a great deal of time and good faith effort has been invested in the preparation of this document to ensure the accuracy of the information it contains.

Patent Identification

The reader's attention is called to the possibility that compliance with this document may require the use of one or more inventions covered by patent rights.

By publication of this document, no position is taken with respect to the validity of those claims or any patent rights in connection therewith. The patent holders so far identified have, we believe, filed statements of willingness to grant licenses under those rights on reasonable and nondiscriminatory terms and conditions to applicants desiring to obtain such licenses.

The following patent holders and patents have been identified in accordance with the TIA intellectual property rights policy:

- No patents have been identified

TIA shall not be responsible for identifying patents for which licenses may be required by this document or for conducting inquiries into the legal validity or scope of those patents that are brought to its attention.

Limited Use Only

Table of Contents

1	Introduction.....	1
1.1	Scope	2
1.2	References	4
1.2.1	Normative References	4
1.2.2	Informative References	4
1.3	Acronyms and Abbreviations	4
1.4	Definitions.....	5
1.5	Conventions	6
1.5.1	Aliases.....	6
1.5.2	Symbols.....	6
2	Conventional Channels.....	7
2.1	Organization of Channels	7
2.1.1	Formats (Informative)	7
2.1.1.1	Voice Format (Informative)	7
2.1.1.2	Data Format (Informative)	7
2.1.1.3	Supplementary Data Format (Informative)	7
2.2	Channel Discipline for the FNE.....	7
2.3	Channel Discipline for the SU.....	8
2.4	Call Termination	8
2.4.1	FNE Transmission Termination.....	8
2.5	Network Access Code and Repeater Addressing.....	9
2.6	Status Symbols.....	11
3	Repeater Operation	13
3.1	AFC Operation.....	15
4	Channel Access Procedures	16
4.1	Channel Access for Voice Transmission	16
4.2	Transmission of the Conventional TSBK	17
4.2.1	Non Emergency Alarm SD Requests	17
4.2.2	Emergency Alarm SD Requests.....	18
4.2.3	Transmission of SD Acknowledgements	20
4.3	FNE Transmission of the Conventional TSBK.....	21
4.3.1	FNE Transmission of Non Emergency Alarm SD Requests	21
4.3.2	FNE Transmission of SD Acknowledgements	21
5	Conventional Registration Procedures	22
5.1	Conventional Unit and Group Addressing	22
6	Voice Call Procedures.....	23
6.1	Routine Group Call Procedures.....	24
6.1.1	SU to Group Voice Transmission	24
6.1.1.1	Originating SU Actions	24
6.1.1.2	FNE Response	25

6.1.1.3	Receiving Subscriber Unit Actions	26
6.1.1.4	Call Termination.....	28
6.1.1.4.1	FNE Response	28
6.1.1.4.2	Receiving SU Actions.....	28
6.1.2	FNE to Group Voice Transmission	28
6.2	Emergency Call Procedure	28
6.2.1	SU to Group Voice Emergency Call Transmission	29
6.2.1.1	FNE Response to Emergency Call Group Voice Transmission	29
6.2.1.2	Receiving Subscriber Unit Actions.....	29
6.2.1.3	Call Termination.....	29
6.2.1.3.1	FNE Response	29
6.2.1.3.2	Receiving SU Actions.....	30
6.2.2	FNE to Group Voice Emergency Call Transmission	30
6.3	All Call (System Wide Call)	30
6.4	Unit-to-Unit Call Procedure	30
6.4.1	SU Sourced Unit-to-Unit Voice Transmission	30
6.4.1.1	Originating SU Actions.....	30
6.4.1.2	FNE Response	31
6.4.1.3	Receiving Subscriber Unit Actions.....	31
6.4.1.4	Call Termination.....	32
6.4.1.4.1	FNE Response	32
6.4.1.4.2	Receiving SU Actions.....	32
6.4.2	FNE Sourced Unit-to-Unit Voice Transmission	32
6.5	Unit to PSTN Call Procedure	32
6.5.1	Unit to PSTN Service Request.....	33
6.5.1.1	FNE Response	33
6.5.1.2	Subscriber Unit Actions.....	33
6.5.2	Traffic Channel Maintenance	34
6.5.3	Call Termination.....	34
6.5.3.1	Subscriber Unit Actions.....	34
6.5.3.2	FNE Actions.....	34
6.6	PSTN to Group Call Procedure.....	34
6.6.1	FNE Action.....	35
6.6.2	SU Response.....	35
6.6.3	Conventional Channel Maintenance	35
6.6.4	Call Termination.....	35
6.7	PSTN to Unit Call Procedure	35
6.7.1	PSTN to Unit Service Request.....	35
6.7.1.1	Availability Check on Called Party	35
6.7.2	Traffic Channel Maintenance	36
6.7.3	Call Termination.....	36
6.7.3.1	Subscriber Unit Actions.....	36
6.7.3.2	FNE Actions.....	36
7	Conventional Packet Data Procedures	38
7.1	Channel Access for Packet Data Transmission	38
7.1.1	RECEIVE IDLE State	39

7.1.2	RECEIVE FS AND NAC State.....	39
7.1.3	FS AND NAC SEEK State.....	39
7.1.4	WAIT RANDOM TIME T_SHORT State.....	40
7.1.5	WAIT FOR END OF MICRO-SLOT State.....	40
7.1.6	RANDOM BACKOFF DELAY State.....	40
7.1.7	TRANSMIT PACKET State	40
7.1.8	Channel Access Parameters	40
7.2	Receiver NAC Qualification	41
7.3	Packet Data Conveyance	41
8	Supplementary Data Procedures	42
8.1	Emergency Alarm Procedures.....	42
8.1.1	Overview	42
8.1.2	Emergency Alarm Request.....	43
8.1.3	Emergency Alarm Cancellation	44
8.2	Call Alert Procedures.....	44
8.2.1	Initial Call Alert Sourced by a Mobile SU	45
8.2.1.1	Mobile Subscriber Unit Actions.....	45
8.2.1.2	FNE Response	45
8.2.1.3	Initiating Mobile Subscriber Unit Actions	45
8.2.1.4	Target Subscriber Unit Actions	45
8.2.1.5	FNE Response to ACK_RSP_FNE Sourced by an SU	45
8.2.2	Initial Call Alert Sourced by the FNE	46
8.2.2.1	FNE Actions	46
8.2.2.2	Target Subscriber Unit Actions	46
8.2.2.3	FNE Response to ACK_RSP_FNE Sourced by an SU	46
8.3	Radio Check Procedures.....	46
8.3.1	Initial Radio Check Sourced by the FNE	47
8.3.1.1	FNE Actions	47
8.3.1.2	Target Subscriber Unit Actions	47
8.3.2	Initial Radio Check Sourced by a Mobile SU	47
8.3.2.1	Initiating Subscriber Actions	47
8.3.2.2	Receiving FNE Actions.....	48
8.3.2.3	Target Subscriber Unit Actions	48
8.4	Radio Inhibit Procedures	48
8.4.1	FNE Actions	49
8.4.2	Target Subscriber Unit Actions	49
8.5	Radio Uninhibit Procedures	50
8.5.1	FNE Actions	50
8.5.2	Target Subscriber Unit Actions	50
8.6	Message Update Procedures	51
8.6.1	Initial Message Update Sourced by a Mobile SU	51
8.6.1.1	Mobile Subscriber Unit Actions.....	51
8.6.1.2	FNE Actions	51
8.6.1.3	Initiating Mobile Subscriber Unit Actions	51
8.6.1.4	Target Subscriber Unit Actions	52
8.6.1.5	FNE Response to ACK_RSP_FNE Sourced by an SU	52

8.6.2	Initial Message Update Sourced by the FNE	52
8.6.2.1	FNE Actions	52
8.6.2.2	Target Subscriber Unit Actions	52
8.6.2.3	FNE Response to ACK_RSP_FNE sourced by an SU	53
8.7	Status Update Procedures	53
8.7.1	Initial Status Update Sourced by a Mobile SU	53
8.7.1.1	Mobile Subscriber Unit Actions	53
8.7.1.2	FNE Actions	53
8.7.1.3	Initiating Mobile Subscriber Unit Actions	54
8.7.1.4	Target Subscriber Unit Actions	54
8.7.1.5	FNE Response to ACK_RSP_FNE Sourced by an SU	54
8.7.2	Initial Status Update Sourced by the FNE	54
8.7.2.1	FNE Actions	54
8.7.2.2	Target Subscriber Unit Actions	55
8.7.2.3	FNE Response to ACK_RSP_FNE Sourced by an SU	55
8.8	Status Query Procedures	55
8.8.1	Initial Status Query Sourced by the FNE	55
8.8.1.1	FNE Actions	55
8.8.1.2	Target Subscriber Unit Actions	56
8.8.1.3	FNE Response to STS_Q_RSP from an SU	56
8.8.2	Initial Status Query Sourced by a Mobile SU	56
8.8.2.1	Subscriber Unit Actions	56
8.8.2.2	FNE Response to STS_Q_REQ Sourced by SU	56
8.8.2.3	Initiating Subscriber Unit Operation	56
8.8.2.4	Target Subscriber Unit Actions	56
8.8.2.5	FNE Response to STS_Q_RSP from an SU	57
8.9	Radio Unit Monitor Procedures	57
8.9.1	Initial Radio Unit Monitor Sourced by the FNE	57
8.9.1.1	FNE Actions	57
8.9.1.2	Target Subscriber Unit Actions	57
8.9.1.3	FNE Response to ACK_RSP_FNE Sourced by an SU	58
8.9.2	Initial Radio Unit Monitor Sourced by a Mobile SU	58
8.9.2.1	Subscriber Unit Actions	58
8.9.2.2	FNE Actions	58
8.9.2.3	Initiating Mobile Subscriber Unit Actions	58
8.9.2.4	Target Subscriber Unit Actions	59
8.9.2.5	FNE Response to ACK_RSP_FNE Sourced by an SU	59
8.10	Alert Tones	59
9	Protection	60
10	Parameter Values	61
Annex A	Inbound Signaling Packet (ISP) Names (Informative)	63
Annex B	Outbound Signaling Packet (OSP) Names (Informative)	64
Annex C	Call Sequence Diagrams (Informative)	65
Annex D	Transmission Preambles (Informative)	73

List of Figures

Figure 1 – Conventional LMR System Diagram.....	3
Figure 2 – Separated Repeaters Serving a Wide Area	9
Figure 3 – Co-Channel User Examples	10
Figure 4 – Channel Status Example	11
Figure 5 – Repeater Configurations	13
Figure 6 – Slotted Status Symbols Example	14
Figure 7 – Assertion of a Status Symbol.....	15
Figure 8 – Conventional Voice Channel Access State Diagram	16
Figure 9 – Non Emergency Alarm Request Transmission State Diagram	17
Figure 10 – Transmission of an Emergency Alarm TSBK State Diagram	19
Figure 11 – Transmission of an SD Acknowledgement TSBK State Diagram	20
Figure 12 – Conventional Data Channel Access State Diagram.....	38
Figure 13 – A Typical Call Sequence Diagram for Unit to Group Voice Call, Unit to Group Emergency Call, and Unit to Unit Voice Call.....	65
Figure 14 – A Typical Call Sequence Diagram for Unit PSTN Voice Call	66
Figure 15 – A Typical Call Sequence Diagram for PSTN to Group Voice Call.....	67
Figure 16 – Typical Call Sequence Diagram for PSTN to Unit Voice Call.....	68
Figure 17 – Typical Call Sequence Diagram for Emergency Alarm	68
Figure 18 – A Typical Call Sequence Diagram for Radio Inhibit (and Radio Check and Radio Uninhibit).....	69
Figure 19 – Typical Call Sequence Diagram for Call Alert Sourced by an SU	69
Figure 20 – Typical Call Sequence Diagram for Call Alert Sourced by the FNE	69
Figure 21 – Typical Call Sequence Diagram for Message Update Sourced by an SU and Directed to a UID (Status Update is similar).....	70
Figure 22 – Typical Call Sequence Diagram for Message Update Sourced by an SU and Directed to a GID (Status Update is similar).....	70
Figure 23 – Typical Call Sequence Diagram for Message Update Sourced by the FNE (Status Update is similar).....	70
Figure 24 – Typical Call Sequence Diagram for Status Query Sourced by the FNE	71
Figure 25 – Typical Call Sequence Diagram for Status Query Sourced by an SU...	71
Figure 26 – Typical Call Sequence Diagram for Radio Monitor Sourced by the FNE.....	71
Figure 27 – Typical Call Sequence Diagram for Radio Monitor Sourced by an SU .	72

List of Tables

Table 1	– Conventional Companion Documents	1
Table 2	– Encrypted Voice Message Conditions.....	25
Table 3	– Receiver Unmute Conditions for Group Voice Transmissions.....	27
Table 4	– Receiver Unmute Conditions for Unit to Unit Voice Transmissions.....	32
Table 5	– Recommended Channel Access Procedure Parameters	41

Limited Use Only

Revision History

Version	Date	Description
Interim Standard	12/13/1993	Basic conventional procedures for voice and data operation
Original	10/02/2003	Revised to explain AFC operation and AFC unlock condition; revised with corrections for address fields and for monitor function; changed messages to packets
Revision A	02/26/2010	Major update to include more normative procedures; re-formatted to a format similar to Trunking Procedures; updated based on CAPPTG comment resolutions; updated based on TR-8.15 comment resolutions; added general FNE address for supplementary data; modified section 8.4.1, paragraph 3, to replace “shall” with “may” because the procedure is not required based on the first “may” in the paragraph; modified section 5.1 to add the requirement that acknowledgements by the FNE to supplementary data messages targeted to a group shall use the General FNE Address as the source address; changed Annex E to normative
Revision B	07/21/2015	Revised to update the procedures related to Packet Data; moved the relevant Packet Data text back into the main body of the document; removed the Packet Data Logical Link Control Procedures and referenced them in the new standalone document; updated the Telephone Interconnect procedures per the latest published document; updated references

This page intentionally left blank

Limited Use Only

1 Introduction

This document describes procedures for both conventional subscriber units (mobile portable and fixed) and the conventional LMR system on which the subscriber units operate. These procedures are sufficient for basic operation of conventional LMR systems. They are not intended to describe operation of trunked systems which have additional operating procedures defined elsewhere. These procedures are intended to be interpreted with the *FDMA Common Air Interface* [4] and are not intended to be understood in isolation.

The basic procedures defined in this document include those for transmitting and receiving digital voice on a radio channel, those for transmitting and receiving data packets on a radio channel, those for making interconnect calls to the PSTN, and those for transmitting and receiving conventional supplementary data messages (using Trunked Signaling Data Blocks (TSBK)).

Section 2 describes the organization of conventional channels including the use of NAC codes and status symbols. In order to describe the transmission of voice, data and TSBKs, it is necessary to define addressing concepts for subscriber radios and radio repeaters, and also to define operating procedures for repeaters. These topics are covered in clause 5.1 and section 3 respectively. Section 3.1 describes channel access procedures for conventional voice and conventional TSBKs. Section 6 describes voice procedures for group, unit-to-unit and PSTN calls. Section 7 describes packet data procedures. Section 8 describes supplementary data procedures. Section 9 describes information related to protecting transmissions of subscriber radios.

These procedures are required to permit interoperability, so that subscriber units and infrastructure from different vendors, and used by different agencies, can interoperate.

This document is one of a set of documents defining the TIA-102 conventional protocol. The companion documents are shown in Table 1 below.

Table 1 – Conventional Companion Documents

Designator	Reference	Description
AABF	[2]	details the presentation of Link Control information
AABG	[3]	details the formats of the Conventional Control packets
BAAA	[4]	details FDMA Common Air Interface
BAAD		this standard
BADA	[11]	details Telephone Interconnect for trunked and conventional subscribers
BAEA	[7]	Data Overview and Specification

1.1 Scope

This document specifies the procedures for conventional LMR systems that permit the conveyance of digital voice, data packets, and supplementary data messages over the common air interface, and permit interconnect calls to a PSTN.

This document does not explicitly address any procedures required in support of any non-conventional (e.g., trunking) modes of operation.

A conventional LMR system is one that does not assign radio channels to SUs through the use of a control channel and over-the-air signaling. It is different from a trunking LMR system which includes some means of assigning radios to channels.

A digital conventional LMR system is one which meets the minimum requirements specified in this document. Additional functions or features can be added as required. The operation of these additional functions is not discussed in this document, but it is expected that these are compatible with the operation as described here. Procedures covered in this standard are further restricted to the lower 2 OSI layers of the digital LMR system. Operation of the encryption function, vocoder function, or user data applications is not explicitly defined here. Operation of trunking control is also not included.

There are several different types of conventional LMR systems, depending on whether or not a repeater is used. These are diagrammed in Figure 1 below.

For the purposes of this document, conventional LMR systems can be classified as either repeater LMR systems or direct LMR systems. Repeater LMR systems make use of a full duplex fixed station that is configured so that all of the signals that are received are re-transmitted. Direct LMR systems transmit directly from one unit to another without the assistance of any intervening repeater. Radio units in an LMR system consist of mobile or portable units, and fixed stations. The mobile or portable units are represented as SU elements and the fixed stations are represented as FS elements in Figure 1 below. The communication paths are represented as arrows. They are intended to show a simplex (i.e. one direction only) path. Generally, the SU is capable of operating half duplex, which is to say that it can transmit or receive, but not both at the same time. The FS is always full duplex. Full duplex operation for portables or mobiles is not discussed for a conventional LMR system.

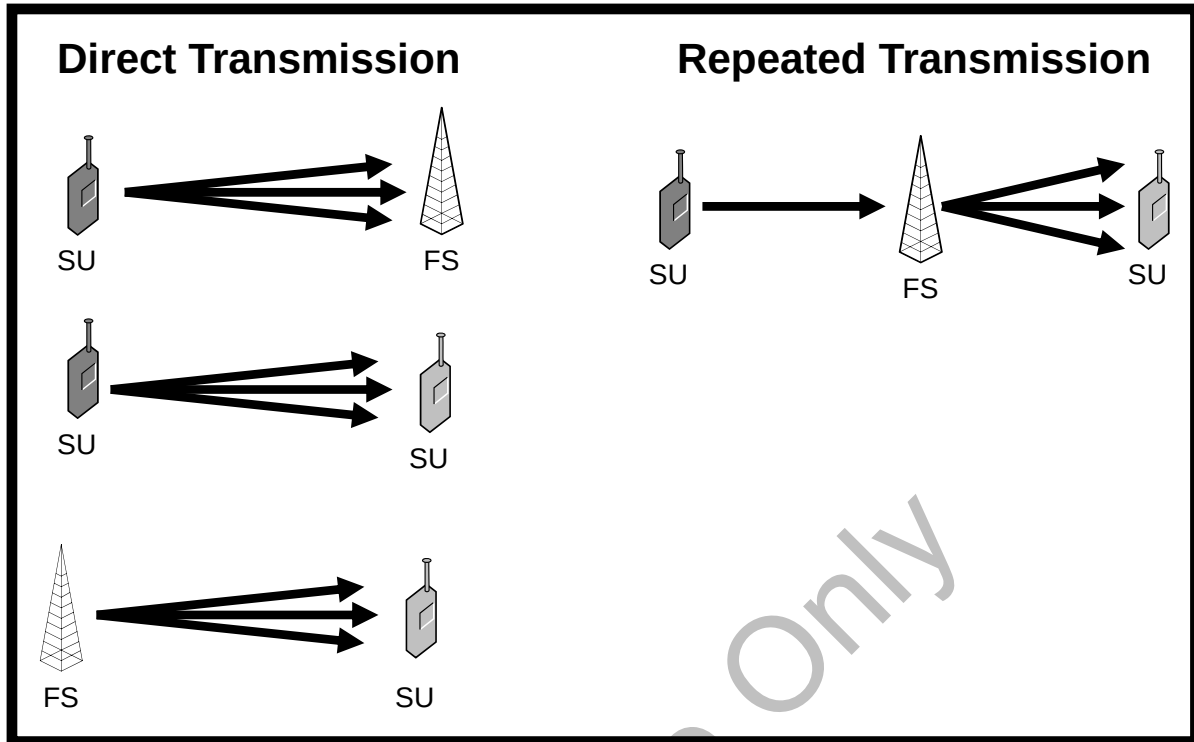


Figure 1 – Conventional LMR System Diagram

SU and FS addressing is covered in section 2 below. The SU and FS also generate Status Symbols, and this subject is covered in clause 2.6 below.

From the stand point of the originating unit, the message is broadcast to all other radio units that are within range of the signal. Those units that are within range either choose to unmute and listen to the signal, or remain muted.

For the purposes of conventional direct LMR systems, either an SU or an FS can be the origination or destination unit. Three representative cases are given in Figure 1 above. Multiple arrows from the originating unit are intended to show communication paths to multiple destination units, of which only a single destination unit is shown.

For conventional repeater LMR systems, the FS is the repeater, and SUs are the origination and destination units. Multiple destination units can listen to the repeater transmission, but only the repeater is configured to listen to the originating unit. The operation of the repeater is a special case of operation of the units in direct mode.

When this document refers to the FNE, it is assumed that it contains at least one FS and possibly consoles.

1.2 References

The appearance of “Project 25” in references below indicates the TIA document has been adopted by the Project 25 Steering Committee as part of its Project 25 standard, i.e. “the Standard.”

1.2.1 Normative References

The following documents contain provisions that, through reference in this text, constitute provisions of this document. At the time of publication, the editions indicated were valid. All documents are subject to revision, and parties to agreements based on this document are encouraged to investigate the possibility of applying the most recent editions of the documents indicated below. ANSI and TIA maintain registers of currently valid national standards published by them.

- [1] TIA-102.AAAD-A, Project 25 Digital Land Mobile Radio Block Encryption Protocol, ANSI/TIA, August 2009
- [2] TIA-102.AABF-D, Project 25 Link Control Word Formats and Messages, ANSI/TIA, April 2015
- [3] TIA-102.AABG, Project 25 Conventional Control Messages, TIA, April 2009
- [4] TIA-102.BAAA-A, Project 25 FDMA Common Air Interface, ANSI/TIA, September 2003
- [5] TIA-102.BAAC-C, Project 25 Common Air Interface Reserved Values, ANSI/TIA, April 2011
- [6] TIA-102.BABA-A, Project 25 Vocoder Description, ANSI/TIA, February 2014
- [7] TIA-102.BAEA-B, Project 25 Data Overview and Specification, ANSI/TIA, June 2012
- [8] TIA-102.BAED, Project 25 Packet Data Logical Link Control Procedures, ANSI/TIA, October 2013

1.2.2 Informative References

- [9] TSB-102-B, Project 25 TIA-102 Documentation Suite Overview, TIA, June 2012
- [10] TIA-102.AABD-B, Project 25 Trunking Procedures, TIA, November 2014
- [11] TSB-102.BADA-A, Project 25 Telephone Interconnect Overview (Voice Service), TIA, June 2012

1.3 Acronyms and Abbreviations

AFC	Automatic Frequency Control
ALGID	Algorithm Identifier
CAI	Common Air Interface
CTC	Call Termination Cancellation
CTCSS	Continuous Tone Coded Squelch Systems
ETDU	Extended Terminator Data Unit
FDMA	Frequency Division Multiple Access

FNE	Fixed Network Equipment
FS	Fixed Station
GID	Group Identifier
KID	Key Identifier
LC	Link Control
LMR	Land Mobile Radio
LSD	Low Speed Data
MI	Message Indicator
NAC	Network Access Code
NID	Network Identifier
PSTN	Public Switched Telephone Network
PTT	Push-to-talk
SD	Supplementary Data
SU	Subscriber Unit
TDU	Terminator Data Unit
TSBK	Trunked Signaling Block
UID	Unit ID

1.4 Definitions

ALGID See definition in [1]

Console Wireline Priority

In repeater-based LMR systems with connected wireline consoles, a console and SU are allowed to transmit simultaneously on the same channel. When this happens the console is given priority and captures the repeat path (the station is transmitting the console's audio instead of repeating the SU's inbound audio). Although the station is not repeating the SU's audio, it continues to route the SU's inbound audio to the infrastructure. This means the consoles that are monitoring the resource are able to hear the transmitting console's audio and the transmitting SU's audio at the same time. This functionality is called "Console Wireline Priority" (it is also referred to as Console Priority or Console Takeover).

Fixed Station See definition in [9]

Frame Sync Frame Synchronization pattern to mark the first information bit

General FNE Address

A source or destination address on the air interface for Supplementary Data that represents the General FNE instead of a specific UID. This allows the initiating SU to send supplementary data messages to interested infrastructure entities (like consoles) without knowing their specific address. When infrastructure devices send Supplementary Data they can also use this address as a source address instead of a specific UID.

GID The 16-bit talk group ID used on conventional channels.

Impolite Channel Access

Refers to the method whereby a device wishing to send a message on a channel does so regardless of the “busy” state of the inbound channel. In this way it could step on other active transmissions.

KID See definition in [1]

LMR System See LMR System in [9]

MI See definition in [1]

NID Network Identifier codeword following the Frame Sync

Polite Channel Access

Refers to the method whereby a device wishing to send a message on a channel does so only if it believes the channel is idle. In this way it does not step on other transmissions.

SU See definition in [9]

Talk Group ID

The conventional Talk Group ID is 16 bit number used to identify or address a group of users on conventional channel. The talk group ID is also referred to as a group ID.

TSBK The message format used to send conventional control (supplementary data) messages.

UID The conventional Unit ID is a 24 bit number used to identify or address an SU on a conventional channel. The Unit ID can be significant only on the channel or on multiple channels.

1.5 Conventions

The following conventions are utilized in this document.

1.5.1 Aliases

To help clarify the definitions in this document, short alias names have been used in place of the full name of the inbound and outbound messages and are shown in Annex A and Annex B below.

1.5.2 Symbols

If a number is preceded by a "%", the number is interpreted as a binary representation with the appropriate number of symbols. If a number is preceded by a "\$", the number is interpreted as a hexadecimal representation with the appropriate number of symbols. If a number is presented with neither a "%" or "\$" preceding it, the number is considered as a decimal representation with the appropriate number of symbols. Representative examples follow:

- %1010 is 4-symbol binary for decimal 10
- \$0A is 2-symbol hexadecimal for decimal 10
- 15 is decimal 15

2 Conventional Channels

2.1 Organization of Channels

Each RF site of the FNE can have one or more conventional radio channels for the access of SUs to the FNE. These conventional channels are used for transmission of digitized voice, packetized Data and Supplementary Data (stand alone Trunked Signaling Data Blocks (TSBKs)).

2.1.1 Formats (Informative)

Voice, Packet Data, and Supplementary Data messages are sent over the air through the FDMA traffic channel as data units. Each data unit for FDMA channels is preceded by a Frame Sync and a NID. Frame Sync is used by the receiver to locate the first bit of the data unit. The NID is used for interference protection and to identify the type of data unit.

A Link Control Word, embedded in a Logical Data Unit 1 of a voice message or in a Terminator Data Unit as specified in [4] is used to carry signaling and information on the conventional channel [2].

2.1.1.1 Voice Format (Informative)

When used for voice service, the FDMA traffic channel transmits and receives digitized voice with format conforming to [4].

2.1.1.2 Data Format (Informative)

When used for data service, the FDMA traffic channel transmits and receives data messages with format conforming to [7].

2.1.1.3 Supplementary Data Format (Informative)

When used for supplementary data service, the FDMA traffic channel transmits and receives data messages with format conforming to [3].

2.2 Channel Discipline for the FNE

The FNE may receive messages which conform to [4] (voice), [7] (Packet Data), or [3] (Supplementary Data).

When retransmitting a received message, the FNE shall re-transmit only those messages which contain an appropriate NAC. Further details of NAC validation can be found in the network access code (NAC) procedures in 2.5 below.

The FNE shall monitor its channels while they have active transmissions. If there is no activity on the channel (e.g., the FNE does not receive data units on the channel and no data units are transmitted as a result of other site or wireline dispatch activity) for **TNoAct** seconds, the FNE shall end the transmission. Note that the value of **TNoAct** may be defined based on transmission type and other conditions.

2.3 Channel Discipline for the SU

Polite operation refers to checking for activity on the inbound channel (potentially based on status symbols received on the outbound channel) and transmitting only if there is no activity detected. An SU may operate politely on the channel before beginning to transmit.

2.4 Call Termination

A Call Termination message is sent on a conventional channel at the end of a voice message or end of a series of messages constituting a voice call.

On the inbound path, subscribers shall transmit a Simple Terminator Data Unit as defined in [4] to signify the end of a transmission. A timer may exist in the SU to limit transmit time. It is recommended that the SU send no more than three Simple Terminators at the end of its transmission.

On the outbound path, to terminate a transmission the FNE may send one or more Simple Terminator Data Units or Terminator Data Units with Link Control as defined in [4]. In [4] the specific data unit is denoted as the LC_CALL_TRM_CAN Link Control. In this document the shortcut ETDU/CTC (Extended Terminator Data Unit / Call Termination Cancellation) is used to denote the data unit. A repeating FNE may convert a received SU's Simple Terminator Data Unit into one or more ETDU/CTC on the repeat path. The FNE shall send target address \$FFFFFFF for group calls and either \$FFFFFFF or one of the individual IDs (source/target) for unit to unit calls. Upon receipt of either a Simple Terminator Data Unit or a ETDU/CTC, SUs shall consider an active transmission on the conventional channel to be over.

2.4.1 FNE Transmission Termination

The FNE shall be able to terminate a transmission by transmitting one or more Call Termination messages (as described in 2.4 above) on the conventional channel:

The FNE shall tear down a transmission under the following conditions:

- An appropriate indication that the call has ended is received from a valid non-radio call participant (e.g. PSTN landline user or console dispatcher); or
- If there is no activity on the allocated channel for **TNoAct**; or
- If a Simple Terminator Data Unit is received from the transmitting SU.

2.5 Network Access Code and Repeater Addressing

To provide increased protection from undesired co-channel interference on control and voice channels, the Network Access Code (NAC) is used. The NAC field contains a 12 bit value as defined in [4] with reserved values as defined in [5]. Specific use of these 12 bits may be defined by the LMR system operator.

Repeaters are not considered to be a subscriber unit in basic conventional LMR systems, and consequently do not possess a Source ID. They can still be addressed with the Network Access Code in the Common Air Interface. This addressing allows multiple repeaters to occupy the same channel, and remain separate and distinct. The NAC allows a coverage area to be serviced by separate repeaters, and it also allows multiple repeaters to service multiple LMR systems with overlapping coverage areas. These cases are depicted in Figure 2 and Figure 3 below.

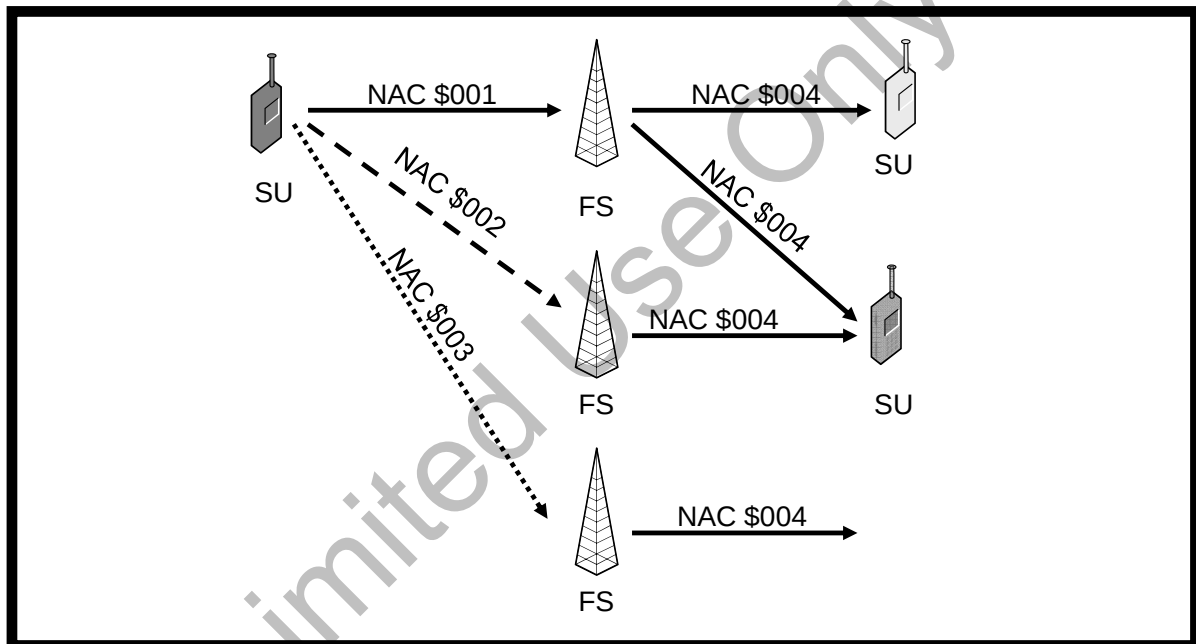


Figure 2 – Separated Repeaters Serving a Wide Area

Figure 2 above shows three repeaters with each repeater using a different NAC on the inbound path and the same NAC on the outbound path. The repeaters can be tuned so that all three inbound paths use the same frequency and all three outbound paths use another common frequency. A subscriber unit can select the NAC based on its proximity to a repeater. For example, within the coverage area of the top repeater, it would use the NAC value \$001. The repeater then re-transmits the message with the NAC value \$004. The subscriber receivers are programmed to receive the NAC value \$004, and in the figure both destination subscribers could receive the message. If the originating unit selected the second repeater, then it might reach a different set of subscribers in a different coverage area. In the figure, only one of the subscribers is reached. The third repeater reaches none of the subscribers shown.

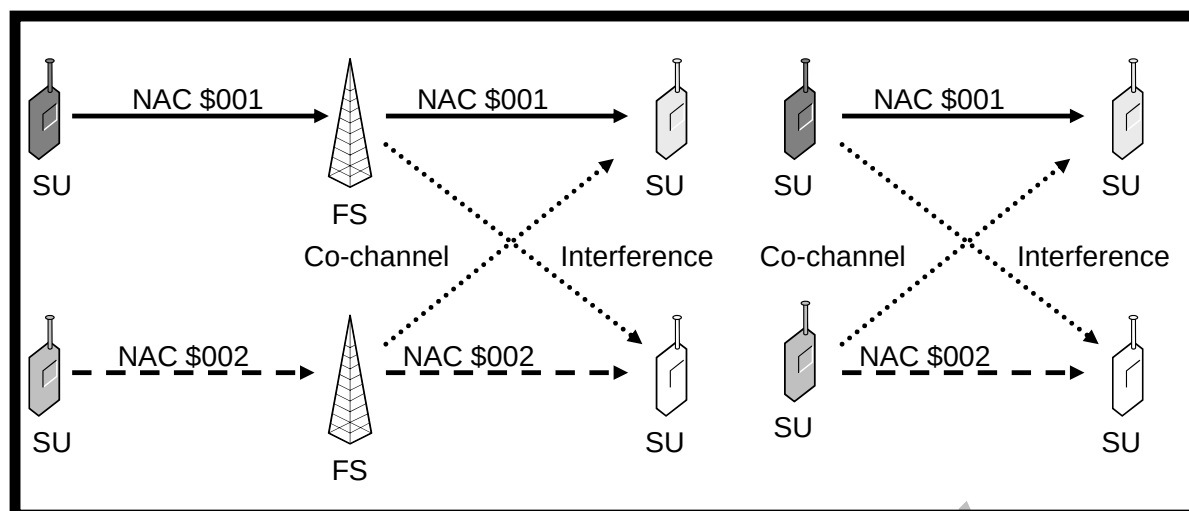


Figure 3 – Co-Channel User Examples

Two examples of co-channel interference are shown in Figure 3 above. In the situation on the left two repeaters are shown sharing a channel, in this case the outbound channel. The subscribers can be programmed to accept messages with the NAC code specified for their particular LMR system and reject co-channel interference from the other LMR system. So long as both repeaters do not transmit simultaneously, the users in each LMR system can share the channel. The situation on the right is very similar except that the channel is shared by subscribers operating in the direct mode.

A repeater shall be assigned at least one NAC value to be used on each of its channels. A repeater may be assigned two NACs for a channel, one for receive and one for transmit. If a repeater is assigned a single NAC value for a channel then it uses the NAC for both receive and transmit. If a repeater is assigned an unreserved NAC value for its receiver it shall only accept and repeat transmissions on the assigned NAC. If a repeater is assigned an unreserved NAC value for a channel's transmit frequency it shall re-transmit received transmissions using the assigned NAC.

There are three reserved NAC values as defined in [5]. NAC value \$293 shall be defined as the default NAC. When NAC \$F7E is assigned to a receiver (a receiver can be an SU or a FS) the receiver shall open on any received NAC. When NAC \$F7F is assigned to a repeater the repeater shall receive and retransmit the received NAC, regardless of its value. Subscriber radios shall not transmit NAC values of \$F7E and \$F7F. These are reserved for use by receivers. Note that \$F7E can be assigned to a SU's receiver and in that case the SU shall transmit a NAC other than \$F7E or \$F7F.

The NAC may be changed by a LMR system operator.

2.6 Status Symbols

The channel status symbols allow a repeater to indicate the status of the inbound channel to subscribers that are waiting for the channel to become available so that they can politely send a voice, packet data or supplementary data message. The status symbol settings of Idle and Busy shall be asserted by a repeater on voice, packet data, and supplementary data messages, and indicate inbound channel activity (Idle = no activity, Busy = activity) for the three call types. An example for packet data is shown in Figure 4 below.

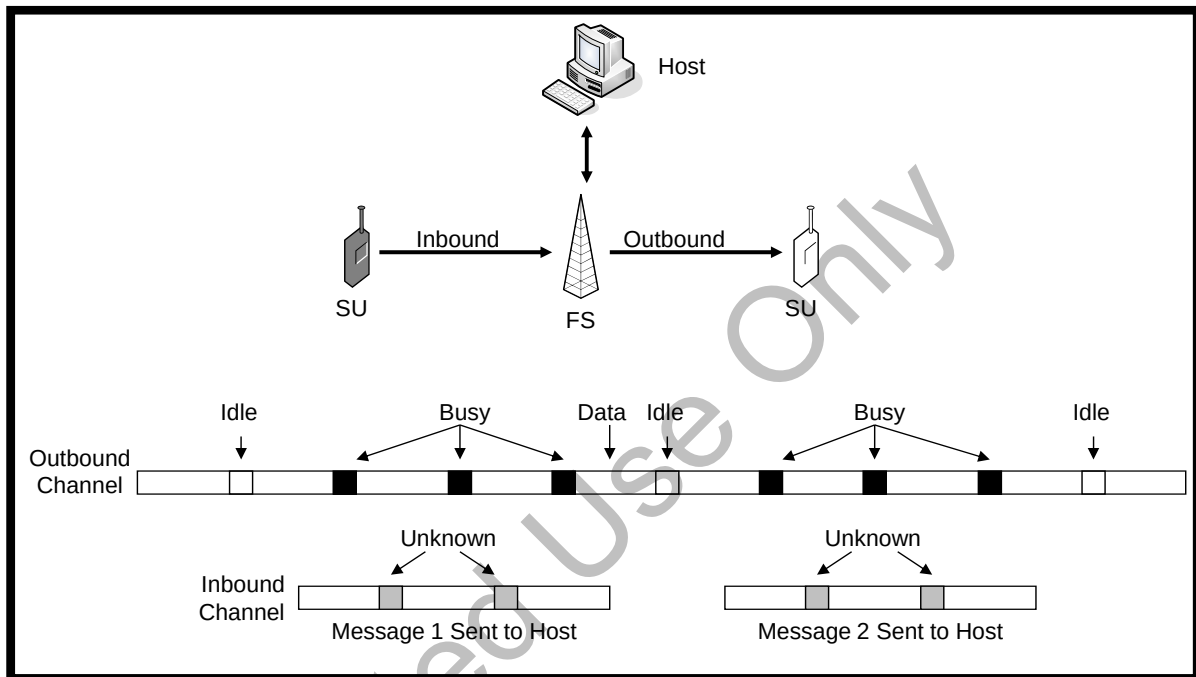


Figure 4 – Channel Status Example

In the example, a repeater is sending a message, or a combination of messages, from a host computer device to a subscriber unit. During the transmission, the inbound channel may be free. If the repeater does not detect the presence of an inbound message, then it sets the embedded status symbol value to Idle. Any subscriber that is waiting to send in a message politely to the host simply waits until the status symbol indicates an Idle condition before it transmits. Once the repeater detects the presence of an inbound message, it sets the status symbol to a Busy value. This can then inhibit other subscribers from transmitting until the inbound message is completed. After the inbound message is completed the repeater clears the status symbol value on the outbound channel to the Idle value so that other subscribers can send messages.

The subscribers shall also embed status symbols in their messages. However, because they are not repeaters, they are unable to indicate the status of the inbound channel. Consequently, they shall set the value of the status symbol to signify an Unknown status. The channel status symbol has 4 possible values. There is one value for Busy, one for Idle, and two values to indicate Unknown status. When the subscriber sends a message on a direct channel, it shall use the Unknown / Direct value for talkaround operation. When the subscriber sends a message inbound to a repeater, it shall use the Unknown value for repeater operation.

Use of status symbols is one method to implement polite operation. If provided by the manufacturer, the LMR system operator may configure their subscribers to ignore status symbols and use other means to implement polite mode (such as looking at the NAC code, sensing carrier, etc.).

Voice transmissions shall be sent polite, except where bypassed based on configuration or local policy. Supplementary Data shall be sent polite except Emergency Alarm which may be sent with a combination of impolite and polite retries. Packet data shall be sent polite.

Limited Use Only

3 Repeater Operation

There are two different configurations for repeaters. In this document they are called the Simple configuration and Fixed Network configuration. These configurations are diagrammed in Figure 5 below. Each configuration treats voice and data messages slightly differently.

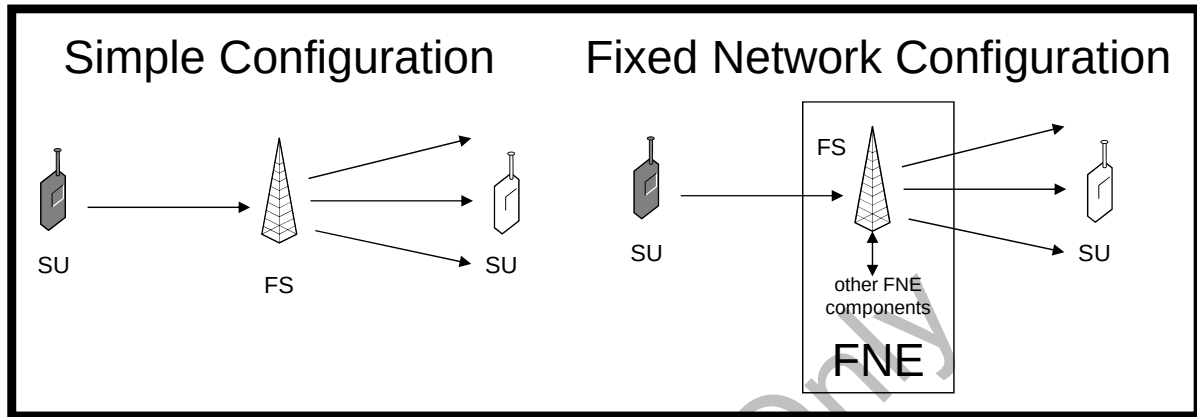


Figure 5 – Repeater Configurations

The operation of a repeater in the simple configuration is obvious. The inbound message shall be received, copied to the transmitter, and transmitted. The receiver of the inbound message may correct errors. The information content of the message shall be unchanged except where noted below. The information content of voice messages includes the coded voice and data, encryption information, and the link control information which includes the group and individual addressing. Note that the repeater is allowed to substitute Link Control information as long as it re-transmits the source Link Control every **NTrafficLC** LC messages, see 6.1.1.2, 6.2.1.1, and 6.4.1.2 below. The information content of data packets includes the contents of all of the blocks. The number of blocks in the data packet is enumerated in the header block as defined in [4].

In the fixed network configuration, when repeating an inbound voice message, the repeater shall copy valid inbound voice messages (e.g. passing the NAC code test) to the outbound path (*with the exception of LC substitution as defined in this section*), and may also copy them to the fixed network, typically to a console device. The receiver of the inbound message may correct errors. An inbound data packet shall be routed based on the form of addressing as defined in [8]. If asymmetric addressing is used, the inbound data packet shall be copied to the fixed network where further processing takes place. If symmetric addressing is used, the inbound data packet may be copied to the outbound path. Data packets originating in the fixed network shall be sent on the outbound path. Voice messages originating in the fixed network shall also be sent on the outbound path. This allows the repeater to transmit and receive either independent data packets, or independent voice messages, at the same time.

A voice or data message also includes status symbols and a Network Access Code (NAC), which are not, in general, copied from the receiver to the transmitter. Instead the repeater may substitute its transmit NAC in place of the received NAC, and it substitutes status information about the inbound channel for outbound status symbols. As defined in 2.5 above, NAC \$F7F causes the station to re-transmit the received NAC, regardless of its value.

Outbound status symbols from a repeater shall be set to the Busy state while the repeater's receiver is receiving an inbound message. The outbound status symbols shall be set to an Idle state when the receiver is not receiving an inbound message. This circumstance is only likely to happen for a repeater in the fixed network configuration. In these LMR systems, the outbound and inbound channels are independent and it is possible to send packets on the inbound channel while a host computer or console is sending an outbound voice, packet data, or supplementary data message.

Status symbols, in general, can be slotted. The operation of slotting simply means that not every status symbol is asserted to be Busy or Idle. Instead, the Busy or Idle indication comes at the end of a slot boundary, and the intervening status symbols on microslot boundaries are set to an Unknown value. A slot consists of N microslots. In the simplest possible case $N=1$, but the standard does not constrain the value of N . Slotting is shown for $N=4$ in Figure 6 below.

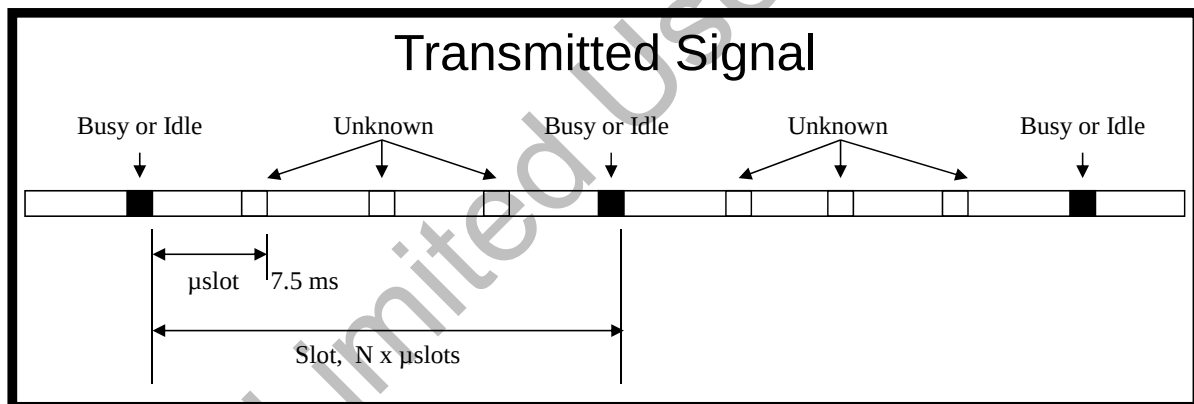


Figure 6 – Slotted Status Symbols Example

The repeater asserts the status symbol at a slot boundary according to activity on the inbound channel. The activity indication may be determined by the detection of carrier and of the Frame Synchronization sequence defined in the Common Air Interface [4]. In some cases this detection may be further qualified by the detection of the Network Access Code in the NID word following the Frame Sync. The case for qualification with the NAC is presented in 2.5 above and is shown in Figure 2 above where more than one repeater shares a channel and is accessed by the specific code. Qualification with the NAC may also be done to reject co-channel interference as shown in Figure 3 above. When the repeater's receiver has detected activity, it shall assert a Busy indication on the next status symbol on the next slot boundary. This is shown in Figure 7 below.

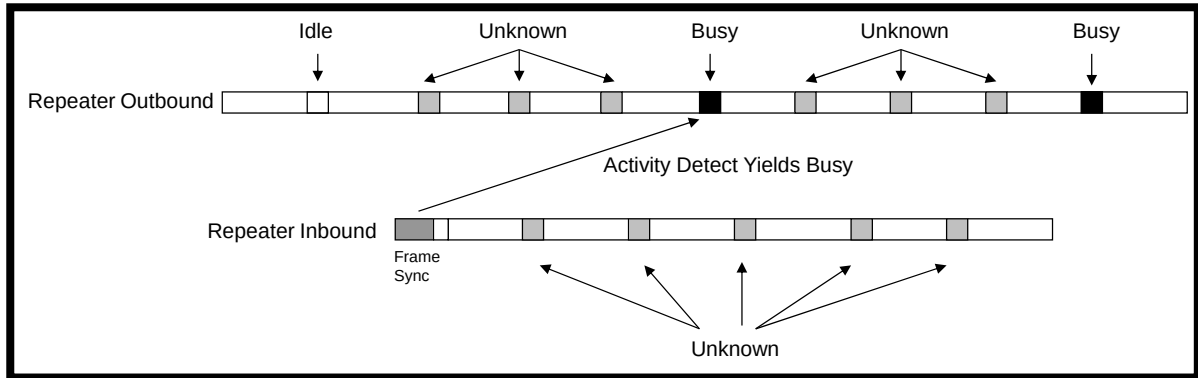


Figure 7 – Assertion of a Status Symbol

Fixed network repeaters may qualify the Frame Sync detect with a NAC, but the most common configuration is likely to be a simple Frame Sync detect. This is because the Frame Sync detect is faster, and allows the repeater to operate with a shorter slot duration.

The choice of slot time is not specified in this document. It is dependent on the operating parameters of the repeater or LMR system.

3.1 AFC Operation

The reference oscillator stability for repeaters and base stations is often better than for subscriber radios. Subscribers may compare the frequency of their local reference oscillator with the carrier frequency from a repeater or base station transmitter, in order to adjust and improve their local reference oscillator. This adjustment is called Automatic Frequency Control (AFC) in this document. AFC operation is anticipated by the FCC regulations for the 769-805 MHz band.

Subscribers may detect a repeater or base station transmission by checking the values of the status symbols on slot boundaries. A repeater or base station transmits Busy or Idle (SUs transmit Unknown) indications on slot boundaries. Repeaters or base stations shall be configured to transmit one or the other of these values on slot boundaries so as to permit subscribers to distinguish repeater or base station transmissions. When a subscriber detects these values, it can average enough data symbols from a transmission to obtain an estimate of the carrier frequency used by the repeater or base station. It can then compare this to the receiver local oscillator to determine any frequency corrections to improve local reference stability.

Subscriber units are in an unlock state after the repeater or base station stops transmitting. AFC locking resumes when a repeater or base station restarts its transmissions.

4 Channel Access Procedures

This section covers the channel access protocols for voice and conventional TSBKs.

4.1 Channel Access for Voice Transmission

This section describes the SU's channel access procedures for voice transmissions. A subscriber transmits on a channel in one of two manners: politely or impolitely. Politely refers to checking the busy status of the inbound channel for voice before transmitting and is the method most commonly used. Impolitely refers to bypassing the channel status check and immediately transmitting.

An SU shall have the ability to use status symbols to determine the busy status of the inbound channel when transmitting voice politely. An SU may be configured to use other means (other than status symbols) to determine the busy status of the inbound channel.

An SU may be configured to transmit voice impolitely for high priority transmissions such as Emergency Call. An SU shall transmit impolitely when the channel is configured for direct mode transmission.

The SU voice transmit procedures are implemented to organize inbound transmissions in a manner which expedites the start of voice transmissions, the completion of voice transactions in progress, and reduces the potential for collisions between contending SUs. These procedures are described below, with reference to Figure 8 below.

For voice transmission the subscriber may be configured to always check the channel busy status, never check the busy status, or check/not check depending on the service being initiated. When the SU has a voice message to transmit it starts in the *Check If Slot Idle* state and determines the availability of the channel using a combination of its configuration and the actual state of the channel. If it determines the channel is Idle, or that it is not Idle but it can ignore it, then it transmits. Otherwise it informs the user that the channel is not available.

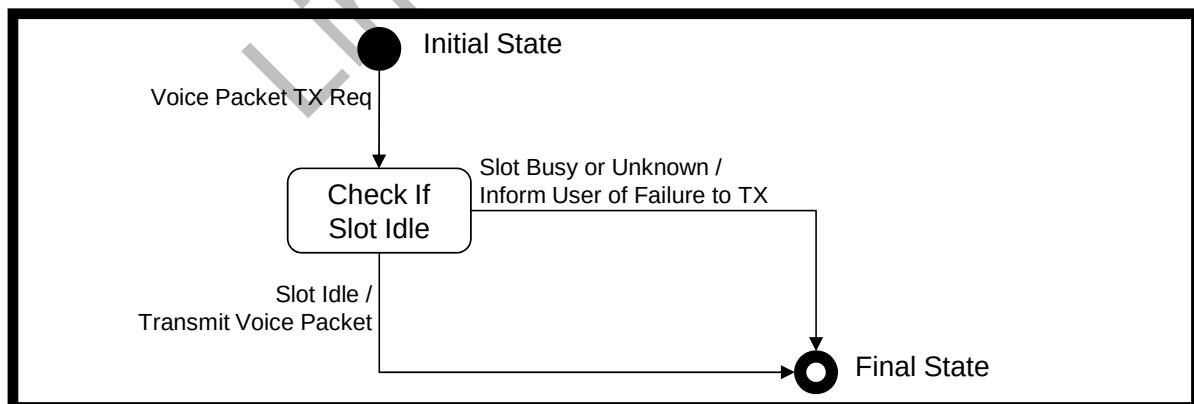


Figure 8 – Conventional Voice Channel Access State Diagram

4.2 Transmission of the Conventional TSBK

This clause covers the transmission of conventional TSBKs (Trunked Signaling Block). Conventional TSBKs are used to transport Supplementary Data (SD) messages. The conventional TSBKs shall be formatted as defined in [3].

There are three different types of SD TSBK transmissions: (1) Non Emergency Alarm SD requests, (2) Emergency Alarm SD requests, and (3) SD acknowledgements. Non-Emergency Alarm SD requests shall be sent polite. Polite transmission of the TSBK refers to the SU ensuring that the inbound channel is idle before transmitting the TSBK. Emergency Alarm requests are sent with a combination of impolite and then polite transmissions. Impolite transmission of the TSBK means the SU transmits the message without checking the state of the inbound channel. Acknowledgements are sent in response to receiving a SD request that requires an acknowledgement. All SD acknowledgements shall be sent polite without retries. The procedures for Non Emergency Alarm SD requests and SD Acknowledgements also apply to FNEs that source SD messages based on internal stimuli (such as a request from a console).

This section describes an approach for TSBK polite channel access methodology based on the detection of status symbols. Although other methods are possible, any approach chosen shall adhere to the normative statements contained herein.

4.2.1 Non Emergency Alarm SD Requests

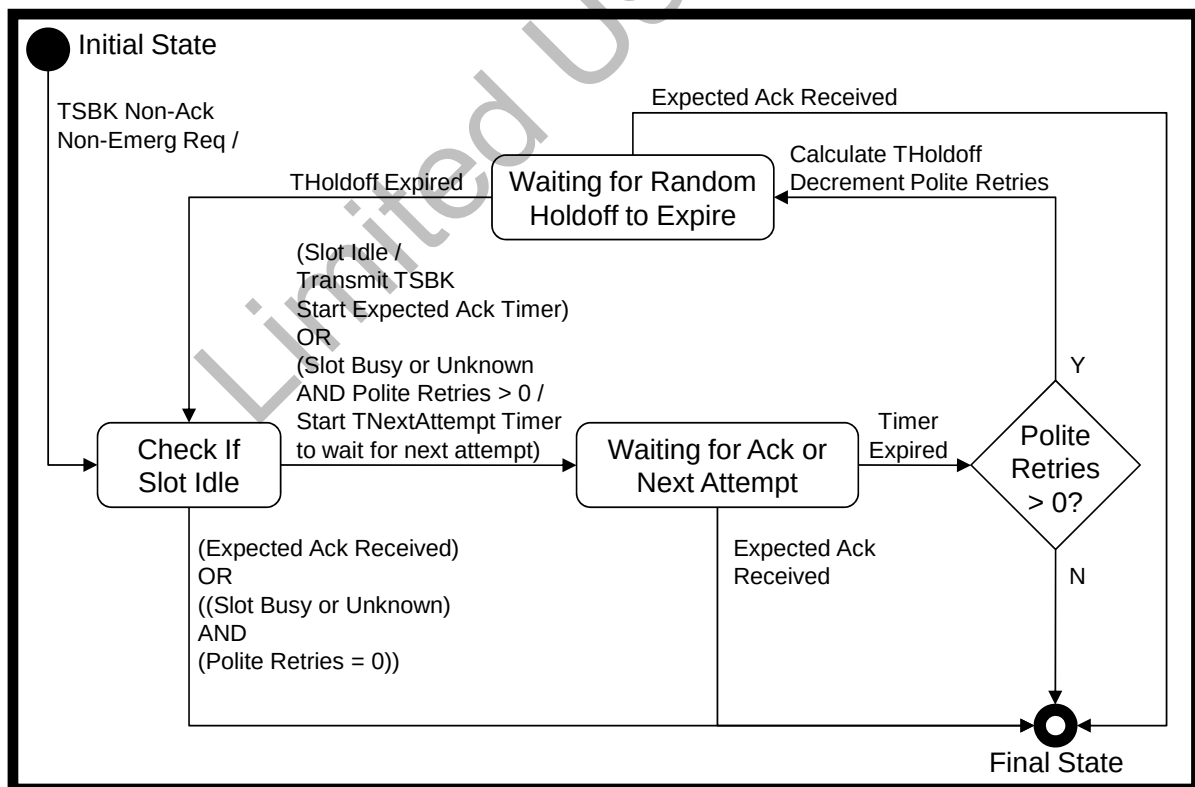


Figure 9 – Non Emergency Alarm Request Transmission State Diagram

Figure 9 above shows an example state transition diagram for the transmission of conventional non Emergency Alarm request TSBK data units. Non Emergency Alarm request data units shall be sent polite. When a user initiates a non Emergency Alarm SD transmit request the subscriber transitions to the *Check If Slot Idle* state and follows polite channel access rules for transmission. In this example the maximum number of attempts is equal to the initial value of Polite Retries plus one.

Upon entry to the *Check If Slot Idle* state, if the channel is idle (no carrier/frame sync being received or status symbols indicates the channel is Idle) the SU transmits the TSBK, starts an Expected Ack timer with a value of **TAck**, and transitions to the *Waiting for Ack or Next Attempt* state. The Expected Ack timer determines when the random Holdoff time and next attempt to send a retry begins. If the slot is not idle and the remaining number of polite retries is greater than 0 then the SU starts the Next Attempt timer with a value of **TNextAttempt** and transitions to the *Waiting for Ack or Next Attempt* state. If the slot is not idle and the remaining number of polite retries is equal to 0 the SU ends the transaction and transitions to the *Final State*. If while in the *Check if Slot Idle* state the SU receives the expected Ack for this message the SU ends this transaction and transitions to the *Final State*.

If, while in the *Waiting for Ack or Next Attempt* state, the timer expires, the SU determines if the number of remaining retries is greater than 0 and if it is it decrements the number of remaining polite retries and starts a random Hold Off timer whose expiration shall determine when the SU tries again to send the TSBK. If no more retries are left the SU ends this transaction and transitions to the *Final State*. If, while in the *Waiting for Ack or Next Attempt* state, the SU receives the expected Ack for this message, the SU ends this transaction and transitions to the *Final State*.

If, while in the *Waiting for Random Hold Off to Expire* state, the SU receives the expected Ack for this message, the SU shall end this state machine and transition to the *Final State*.

4.2.2 Emergency Alarm SD Requests

The Emergency Alarm TSBK shall be sent using a combination of 0 or more impolite attempts and 1 or more polite attempts. A typical combination is 5 impolite attempts followed by 15 polite attempts. Figure 10 below shows the state transition diagram for transmission of an Emergency Alarm TSBK which uses both impolite and polite attempts. The procedure in 4.2.1 above may be used to send Emergency Alarm if only polite attempts are employed.

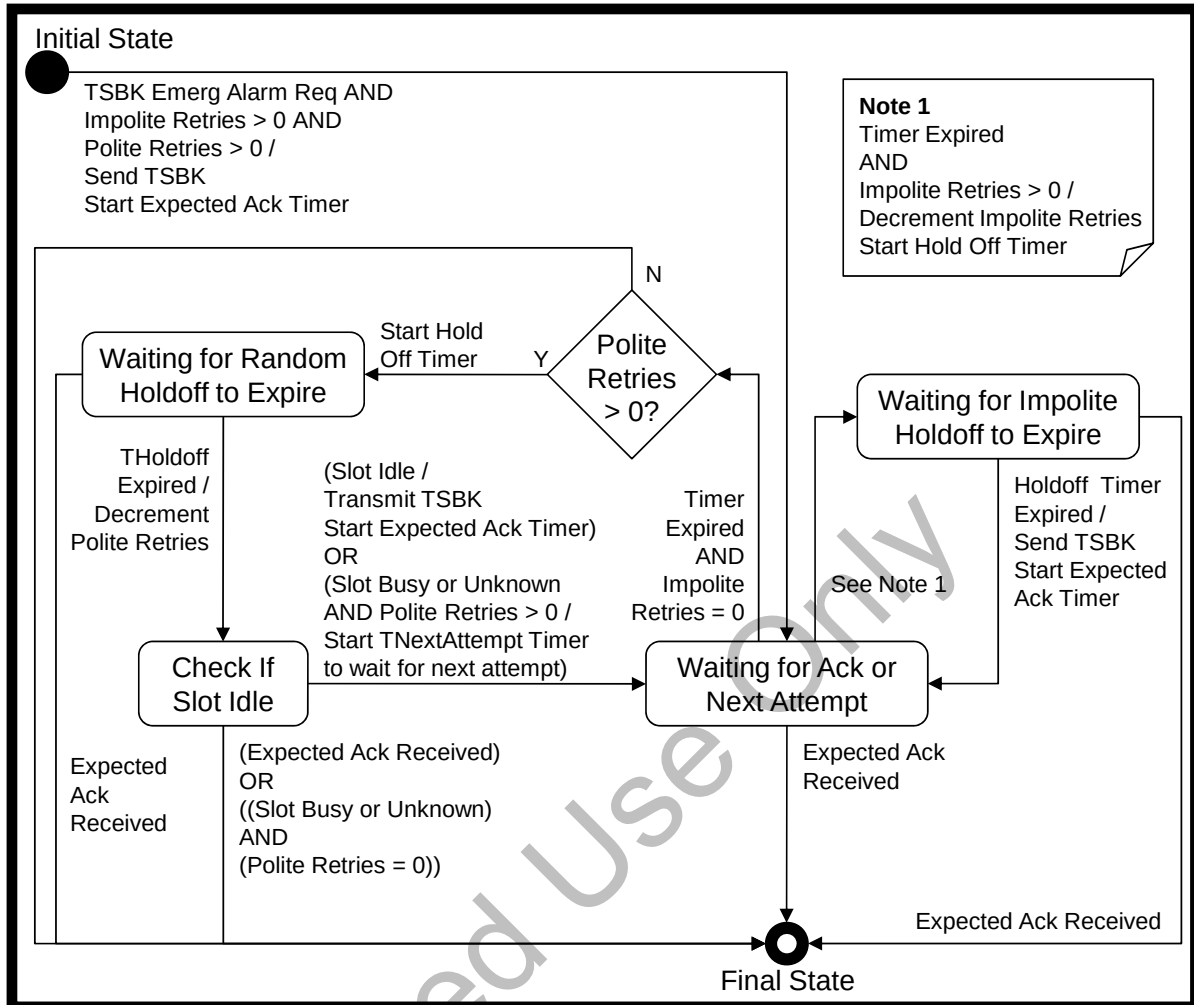


Figure 10– Transmission of an Emergency Alarm TSBK State Diagram

When a user initiates an Emergency Alarm Request transmission and is configured to use impolite attempts to transmit, the SU immediately sends the TSBK impolitely, starts the Expected Ack timer with a value of **TAck**, and transitions to the *Waiting for Ack or Next Attempt* state. The Expected Ack timer determines how long the SU waits for the expected ack before starting the random Holdoff timer which drives when it sends the next retry. If the Expected Ack timer expires and there are impolite retries remaining, the SU starts a random Hold Off timer and transitions to the *Waiting for Impolite Holdoff to Expire* state. If the Expected Ack timer expires and there are no impolite retries remaining but there are polite retries, the SU starts a random Holdoff timer and transitions to the *Waiting for Random Holdoff to Expire* state. If the Expected Ack timer expires and there are neither polite nor impolite retries remaining, the SU ends this transaction and transitions to the final state. If, while in the *Waiting for Ack or Next Attempt* state, the expected acknowledgement for this message is received, the SU ends this transaction and transitions to the final state.

If, while in the *Waiting for Impolite Holdoff to Expire* state, the expected acknowledgement is received, the SU ends this transaction and transitions to the final state. When the random Holdoff timer expires, the SU shall send the next impolite TSBK and transitions back to the *Waiting for Ack or Next Attempt* state. This continues until there are no more impolite retries or the expected ack is received.

If, while in the *Waiting for Random Holdoff to Expire* state, the expected acknowledgement is received, the SU ends this transaction and transitions to the final state. When the random Holdoff timer expires the SU decrements the number of remaining polite retries and transitions to the *Check if Slot Idle* state.

Upon entry to the *Check If Slot Idle* state, if the channel is idle (no carrier/frame sync being received or status symbols indicates the channel is Idle) the SU transmits the TSBK, starts an Expected Ack timer with a value of **TAck**, and transitions to the *Waiting for Ack or Next Attempt* state. The Expected Ack timer determines when the random Holdoff time and next attempt to send a retry begins. If the slot is not idle and the number of polite retries is greater than 0 then the SU starts the Next Attempt timer with a value of **TNextAttempt** and transitions to the *Waiting for Ack or Next Attempt* state. If the slot is not idle and the number of retries is equal to 0 the SU ends the transaction and transitions to the *Final State*. If while in the *Check if Slot Idle* state the SU receives the expected Ack for this message the SU ends this transaction and transitions to the *Final State*.

4.2.3 Transmission of SD Acknowledgements

Figure 11 below depicts the state diagram for the transmission of an SD acknowledgement TSBK.

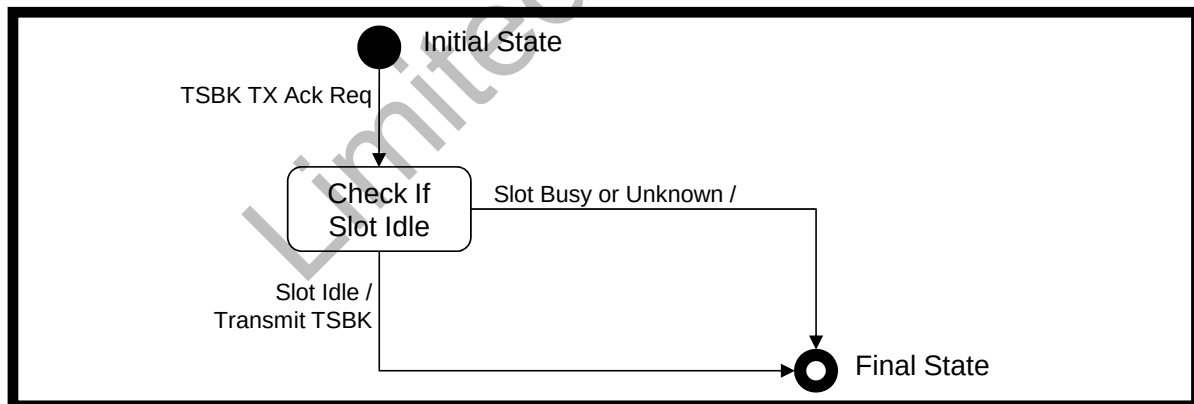


Figure 11– Transmission of an SD Acknowledgement TSBK State Diagram

When an SU initiates the transmission of an acknowledgement TSBK it transitions to the *Check If Slot Idle* state to try to politely transmit one copy of the acknowledgement. If the channel is idle (no carrier/frame sync is being received) or if status symbols indicate that the outbound channel is Idle, the SU shall transmit the TSBK. If the status symbols indicate that the channel is BUSY or UNKNOWN the SU may exit the transaction and rely on the sender to re-try the message that required an acknowledgment. Alternatively, the SU may hold the acknowledgement and send it when the channel is idle.

4.3 FNE Transmission of the Conventional TSBK

This section covers the FNE transmission of conventional TSBKs (Trunked Signaling Block).

There are two different types of SD TSBK transmissions sourced from the FNE: (1) Non Emergency Alarm SD requests and (2) SD acknowledgements. The FNE does not source an Emergency Alarm.

4.3.1 FNE Transmission of Non Emergency Alarm SD Requests

The FNE can implement transmission of non-Emergency Alarm SD requests. These transmissions are sent polite and can follow the principles given for subscribers in 4.2.1 above. Polite mode for the FNE refers to the FNE ensuring that the inbound channel is idle before transmitting the TSBK on the associated outbound channel.

4.3.2 FNE Transmission of SD Acknowledgements

The FNE can implement transmission of SD Acknowledgements. These transmissions are not sent polite and can follow the principles given for subscribers in section 4.2.3 above.

5 Conventional Registration Procedures

This section is mainly a placeholder for Conventional Registration procedures, to be filled in at a later date. Addressing of subscribers is included in this section since it could have an impact on registration procedures.

5.1 Conventional Unit and Group Addressing

A basic conventional LMR system consists of a set of mobile or portable radio units, and perhaps consoles and repeaters which can communicate with each other. Those radio units which can terminate voice or data messages are called subscriber units in the LMR system. A console is a special type of subscriber, sometimes called a Console SU. This definition of subscribers excludes repeaters for the purposes of basic conventional systems. Subscriber units can be addressed with 24-bit Source ID, a 24-bit Destination ID, or with a 16-bit Talk Group ID. [5] describes the values ranges for these fields.

6.1.1.3 and 6.4.1.3 below include additional information regarding selective squelch and talk group IDs and unit IDs.

The Reserved Values document [5] defines Source and Destination IDs in the range of \$989680 to \$FFFFFFE (1,000,000 to 16,777,215) as being for talk group use or other special purposes. The Message Update and Status Update messages (see section 8 below) may make use of this special range of addresses to denote a destination talk group instead of a destination unit ID. When the special range is used to denote a voice talk group, all 16-bit talk group IDs (\$0001 to \$FFFF) shall be mapped to a range of 24-bit talk groups. The 24-bit talk groups shall be the last 65535 Unit IDs that are allowed for talk group use (\$FEF001 to \$FFFEFFF). Thus \$FEF001 denotes talk group \$0001 and so on.

Annex A of the Trunking Procedures document [10] states that UID \$FFFFFFC is reserved for FNE use. This address is also referred to as the General FNE Address. The General FNE Address may be used as a source address for conventional supplementary data sent by the FNE and as a destination address for conventional supplementary data sent by an SU. FNE acknowledgments to messages which used the General FNE address as the destination address shall use the General FNE Address as the source address. FNE acknowledgments to messages which are targeted to a group shall use the General FNE Address as the source address. SU acknowledgments to messages which used the General FNE address as the source address shall use the General FNE Address as the destination address. The General FNE address may be any legal SU ID, and is configured in the FNE and interested SUs. The default General FNE Address for conventional shall be \$FFFFFFC.

Talk Group \$0000 shall only be used in the Header Word (defined in [4]) of a unit-to-unit voice transmission.

6 Voice Call Procedures

This section details the procedures for the beginning and ending of voice transmissions. It includes the possible responses generated by the FNE and the SU actions. In general these procedures apply to both direct and repeat mode calls. The only difference is that direct mode does not include the repeat function described in some of the procedures.

The operation of the subscriber transmitter for voice messages consists of four main cases, with several variations of each case. The four main cases consist of Routine Group Calls, Emergency Group Calls, Individual Calls and PSTN calls. The controls on a radio to initiate these different calls are defined in this section before the call operation is defined.

The subscriber transmitter may have several controls which affect the transmit operations. These controls are sufficient for a subscriber unit to support all of the call types defined here. These controls are as follows.

PTT Switch – The Push-To-Talk switch is activated when the operator wishes to transmit and released when a transmission is finished.

Channel Selector – The Channel Selector is a switch or control that allows the operator of the radio to select a radio's mode of operation. The operational parameters that are selected for the transmitter include the following items.

1. Transmit frequency
2. Transmit Network Access Code
3. Talk Group
4. Other parameters for setting the vocoder and encryption functions. For example, the encryption key could be selected.

Emergency Switch – The Emergency switch is asserted by the radio operator for emergency calling. Once this switch is asserted, the emergency condition remains asserted until it is cleared by a different means, such as asserting the Emergency Switch for an extended period of time.

Numeric Keypad / Display – This allows the radio operator to set numeric parameters, such as the number for a subscriber ID. This is most useful for individual calls.

The different types of calls described here are defined as follows.

Routine Group Call – This is a transmission that is intended to a group of users in the radio system. It is the type of call that is made most often during normal radio operation. Typically, these calls are made when the PTT switch is asserted unless some other special condition exists.

Emergency Group Call – This is a transmission that is intended to a group of users in a LMR system, during an emergency condition. The definition of an emergency condition depends on the LMR system's operators, but it typically signifies an exceptional condition with more urgency for the listeners to the call. These calls are typically made after the Emergency switch is asserted.

Individual Call – This is a transmission which is addressed to a specific individual. The individual subscriber's ID to which the call is directed is called the Destination ID. These calls are made after the Destination ID is entered into the radio.

PSTN Call – This is a transmission between one or more SUs and a user of a phone line. This is also referred to as a PSTN Interconnect call.

The procedures for each of these calls in the transmitter are based on the procedure for the routine group call. Consequently, that call is described first, and then the other calls are described.

6.1 Routine Group Call Procedures

Figure 13 below is a typical call sequence diagram for a group call.

6.1.1 SU to Group Voice Transmission

6.1.1.1 Originating SU Actions

The SU shall comply with the channel access protocol of 4.1 above when making a group voice call by generating a stream of voice frames on the transmit frequency.

The voice frames shall conform to the vocoder description as defined in [6]. The over-the-air transmission shall comply with the FDMA CAI as defined in [4]. If the transmission is encrypted, the voice frames shall be encrypted per the Block Encryption Standard as defined in [1].

The SUs configuration and channel knob selector position determine the value of some of the voice frame parameters and where the values come from is not part of the standard.

The MFID used throughout the transmission shall be filled in as defined in [4].

The Talk Group ID in the Header Word and Link Control field shall adhere to the range of \$0001 to \$FFFF as defined in [5]. The remaining fields in the Header Word (MI, ALGID and KID) shall be set as defined in [1].

While transmitting on the inbound traffic channel, the SU shall fill the LC field with the LC_GRP_V_CH_USR (Group Voice Channel User) message as defined in [2]. Routine Group voice transmissions shall set the Emergency bit in the LC field as defined in [4] (and [2]) to 0. Note that the remaining bits of the Service Options field, as defined in [2] for the LC_GRP_V_CH_USR message may be considered Reserved as defined in [4]. As stated in [4], Reserved bits should be filled in with nulls (zeros). Alternatively, the remaining bits of the Service Options field may be filled in as defined in [2]. As defined in [5] the LC field for Group voice transmissions shall utilize a source unit ID in the range of 1 to 9,999,999.

If an LMR system operator does not want to use talk groups on a channel the operator may configure the subscribers and FNE (for example, a console) that access the channel to operate using the Unaddressed Call mode. Unaddressed Call is a mode of operation that does not use the talk group ID to drive squelch operation and operates in the same manner as analog FM conventional channels with respect to squelch modes. To avoid confusion on the channel, when in Unaddressed Call Mode, the SU and FNE should use NAC codes or similar methods (for example Secure Keys) to segment users in a similar manner that CTCSS codes are used on analog FM channels. It is recommended that when in the Unaddressed Call Mode the SU and FNE use the default talk group ID value of \$0001. Subscribers and FNE may also be configured to use any other valid talk group ID and not squelch based on talk groups. Note that simply configuring the SU or FNE to use talk group \$0001 does not automatically mean it is operating in Unaddressed Call Mode. It is allowable to squelch on talk groups and use talk group \$0001.

The contents of Link Control and LSD fields may also be encrypted if the voice information is encrypted in accordance with Table 2 below.

Table 2 – Encrypted Voice Message Conditions

Fields Encrypted			
Voice	LSD	LC	Conditions
Clear	Clear	Clear	Allowed, LC shall use unencrypted format
Clear	Clear	Encrypted	NOT ALLOWED
Clear	Encrypted	Clear	NOT ALLOWED
Clear	Encrypted	Encrypted	NOT ALLOWED
Encrypted	Clear	Clear	NOT ALLOWED
Encrypted	Clear	Encrypted	NOT ALLOWED
Encrypted	Encrypted	Clear	Allowed, LC shall use unencrypted format
Encrypted	Encrypted	Encrypted	Allowed, LC shall use encrypted format

The fields in the ESYNC Word (MI, ALGID and KID) shall be set as defined in [1] and [5].

6.1.1.2 FNE Response

The following actions are valid responses from the FNE to a group voice transmission, though an FNE need not perform all of these actions:

- (Repeat Mode only) The FNE may transparently repeat the group voice transmission as defined in section 3 above. The FNE may also take into account console wireline priority when determining whether to repeat the transmission.

While transmitting on the outbound channel in repeat mode, the FNE shall send the first LC_GRP_V_CH_USR message received and then repeat subsequent LC_GRP_V_CH_USR messages at least once every **NTrafficLC** messages. The source address shall reflect the currently active source address and the group address shall reflect the currently active working group. The emergency bit in the LC shall reflect the setting of the currently active transmitting SU.

6.1.1.3 Receiving Subscriber Unit Actions

Depending on its squelch setting, a subscriber monitoring the channel may display received call information and play out the audio associated with the received transmission. The following text explains how an SU determines if the received call should be presented to the user.

For receive purposes on a specific channel, every subscriber unit shall be a member of at least one conventional talk group and may be a member of more than one talk group. An SU shall always have one talk group pre-selected for its own transmissions on the channel. This talk group may be overridden by operations such as hang time as described later in this section. Any subscriber unit may be a member of any talk group, and any talk group may contain any number of subscriber units. Talk Groups may be used to segment users into groups which use the same channel but don't hear each other's conversations. Every unit on a channel shall consider itself a member of the special talk group ID value \$FFFF, the All Call talk group. Depending on its squelch mode, the receiving subscriber's currently selected talk group(s) may be used as a filter to determine whether to unsquelch for the group voice transmission.

The operation of the subscriber receiver for voice messages consists of three main cases, with variations that depend on the receiver's operation. The three main cases are called Squelch conditions in this document. They are Monitor, Normal Squelch, and Selective Squelch.

As in the case of the transmitter, the receiver operation is affected by the current settings of the subscriber (which may have been chosen using the selector knob):

- Receive frequency
- Receiver Network Access Code
- Talk Group(s)
- Other parameters for setting the vocoder and encryption functions. The encryption function operation is particularly significant to the receiver. This is covered in the encryption protocol description.

An additional radio control which affects a receiver is a Monitor switch. This allows the operator of the radio to disable normal or selective squelch of the receiver so that the operator can hear any signs of voice activity. This is useful for avoiding collisions on the channel between voice users.

The different types of squelch operation described here are defined as follows.

Monitor – This enables the receiver to unmute on any recognizable voice signal. Selective muting based on the network access code, talk group ID, or Destination ID is not performed. This is analogous to monitor mode in analog receivers. This is normally activated with the Monitor switch.

Normal Squelch – This enables the receiver to unmute on any voice signal which has the correct network access code. Voice messages from co-channel users which are using different access codes shall be muted.

Selective Squelch – This mutes all voice traffic except that which is explicitly addressed to the unit, addressed to the unit's talk group, or addressed to all units. Messages sent to the talk group ID or Destination ID of the receiver or to the All Call talk group \$FFFF, as well as the correct network access code, shall be received.

An SU shall support the Monitor, Normal Squelch, and Selective Squelch modes.

The details of the procedure in the receiver for implementing these different squelch types are not covered here. This is because the receivers should be designed to tolerate fading and interference while decoding the message and the solution to this problem is likely to vary from one design to another. In general, however, the receiver is supposed to unmute if a specific subset of attributes is decoded in the voice message. These attributes include the Network Access Code (NAC), the Manufacturer's ID (MFID), the encryption information, the Talk Group ID (TGID), and the Destination ID. The conditions for unmuting for a normal Group Voice transmission are given in Table 3 below for the three squelch settings. Note that a receiver may operate as if it is in Monitor mode (and not support the Normal or Selective squelch mode) when configured with a receive NAC of \$F7E.

Table 3 – Receiver Unmute Conditions for Group Voice Transmissions

Attribute	Monitor	Normal Squelch	Selective Squelch
NAC	Don't Care	= Receiver NAC or Don't Care if Receiver NAC is \$F7E	= Receiver NAC or Don't Care if Receiver NAC is \$F7E
TGID	Don't Care	Don't Care	= A specific TGID that is assigned to the Receiver or ALL Call TGID \$FFFF
Encryption	Don't Care	Decryption is Possible	Decryption is Possible

The normal squelch mode is the equivalent of Unaddressed Call mode. The talk group ID is not used in evaluating whether to unsquelch for the transmission.

The output of the receiver in the monitor mode, when decryption is impossible, may be muted. In general, it should not output harsh sounding audio (the voice cannot be decrypted in this case). The SU shall provide some form of indication to the user that there is activity on the channel even though it cannot decrypt the audio.

Upon responding to a received transmission with its own voice transmission, an SU may employ a hang time associated with the talk group. Hang time is useful when the SU is responding to an All Call or when it is configured to be a member of more than one talk group. If the user causes the SU to transmit back during the hang time window the SU uses the talk group of the last received call. If it transmits back after the hang time window the SU uses its currently chosen talk group. The hang timer restarts upon the end of every received transmission. Thus, a radio set up for talkgroup 0002 transmits with talkgroup 0002 if there is no activity or if it is responding to activity from talkgroup 02. It transmits with talkgroup \$1111 if it receives a transmission on talk group \$1111 and is keyed within the hang time window. It is recommended that subscribers configured with hang time do not respond to All Calls (talk group \$FFFF) with an All Call transmission. An inbound All Call could wreak havoc on an unsuspecting LMR system.

The received source and talk group ID may be displayed for use by the radio operators. When the ID is displayed numerically, it shall be displayed with decimal digits. Instead of a numeric ID, the SU may display a textual phrase which has been mapped to the numeric source or talk group ID.

6.1.1.4 Call Termination

The transmitting subscriber shall terminate its voice transmission by sending a simple Terminator Data Unit (TDU) as defined in [4].

6.1.1.4.1 FNE Response

The following actions are valid responses from the FNE to reception of a simple Terminator Data Unit, though an FNE need not perform all of these actions:

- (Repeat Mode only) The FNE may transparently repeat the simple TDU as defined in 2.4 above, including transmitting more than one copy.
- (Repeat Mode only) The FNE may convert the simple TDU to an Expanded TDU as defined in [2] and transmit one or more on the channel.

6.1.1.4.2 Receiving SU Actions

Upon receiving a simple TDU or Expanded TDU the receiving SU considers the transmission completed and takes actions based on its implementation.

6.1.2 FNE to Group Voice Transmission

This is an FNE sourced transmission, typically at the request of one of the FNE's consoles. The procedures are the same as 6.1.1 above with the originating FNE taking on the role of the originating SU. Note also that the source FNE may also adhere to the procedures described in 6.1.1.4.1 above.

6.2 Emergency Call Procedure

Figure 13 below is a typical call sequence diagram for Emergency Call.

This section describes the Emergency Call group voice procedure which allows an SU to make a voice transmission that is marked as being in the Emergency mode of operation.

6.2.1 SU to Group Voice Emergency Call Transmission

In the case of an Emergency Call group voice transmission the transmitting SU sets the Emergency bit in the LC_GRP_V_CH_USR (Group Voice Channel User) message to 1 as defined in [4] (and [2]). The bit is set based on the device configuration settings and the user having previously set their device into emergency mode. Each device is responsible for setting or clearing their emergency setting. Thus, while some voice transmissions on a channel may have the emergency bit set, others sourced by different SUs on the same channel may not.

Except for the value of the Emergency bit, the remaining procedures are the same as those defined in 6.1.1 above.

6.2.1.1 FNE Response to Emergency Call Group Voice Transmission

The following actions are valid responses from the FNE to an Emergency Call group voice transmission, though an FNE need not perform all of these actions:

- (Repeat Mode Only) The FNE may transparently repeat the Emergency Call transmission as described in section 3 above. The FNE may also take into account console wireline priority when determining whether to repeat the transmission.

While transmitting on the outbound channel in repeat mode only, the FNE shall send the LC_GRP_V_CH_USR message at least once every TTrafficLC LC messages. The source address shall reflect the currently active source address and the group address shall reflect the currently active working group. The emergency bit shall reflect the setting of the currently active transmitting SU.

6.2.1.2 Receiving Subscriber Unit Actions

The receiving subscriber unit actions shall follow those actions described in 6.1.1.3 above with the following additional actions.

An SU that receives an Emergency Call and whose squelch processing indicates that the transmission is to be presented to the user shall provide an indication that the current received transmission is being sent in Emergency mode.

6.2.1.3 Call Termination

The transmitting subscriber shall terminate its voice transmission by sending a simple Terminator Data Unit (TDU) as defined in [4].

The Emergency Call state of the subscriber shall continue until the user takes an action to clear it. Typical actions required to clear the Emergency include holding down the Emergency switch for an extended period of time. The LMR system cannot cancel the internally held Emergency Call state of an SU.

6.2.1.3.1 FNE Response

The FNE response is the same as defined in 6.1.1.4.1 above.

6.2.1.3.2 Receiving SU Actions

The receiving SU's actions are the same as defined in 6.1.1.4.2 above.

6.2.2 FNE to Group Voice Emergency Call Transmission

This is an FNE sourced Emergency Call transmission, typically at the request of one of the FNE's consoles. The procedures are the same as 6.2.1 above with the originating FNE taking on the role of the originating SU. Note also that the source FNE may also adhere to the procedures described in 6.2.1.3.1 above.

6.3 All Call (System Wide Call)

The All Call is a group voice transmission with a talk group ID of \$FFFF. Every SU shall be a member of talk group \$FFFF. The All Call may be initiated by the FNE. Subscribers should not initiate transmission of an All Call. Details of the FNE call setup procedure to route an All Call received or transmitted on a conventional channel to other conventional and/or trunking channels is beyond the scope of this document.

Regardless of its talk group configuration settings, an SU and FNE (console) shall always consider a received transmission using the All Call talk group \$FFFF to be a transmission to one of its assigned talk groups. The All Call transmission audio and relevant embedded control information shall be presented to the user if the other squelch criteria (such as NAC) are met. The SU and FNE (console) may indicate to the user that an All Call is being received.

6.4 Unit-to-Unit Call Procedure

Figure 13 below is a typical call sequence diagram for unit-to-unit call.

6.4.1 SU Sourced Unit-to-Unit Voice Transmission

6.4.1.1 Originating SU Actions

The SU shall comply with the channel access protocol of 4.1 above when making a unit-to-unit voice call by generating a stream of voice frames on the transmit frequency.

The voice frames shall conform to the vocoder description as defined in [6]. The over-the-air transmission shall comply with the FDMA CAI as defined in [4]. If the transmission is encrypted, the voice frames shall be encrypted per the Block Encryption Standard as defined in [1].

The SUs configuration and channel knob selector position determine the value of some of the voice frame parameters and where the values come from is not part of the standard.

The MFID used throughout the transmission shall be set as defined in [4].

The Talk Group ID in the Header Word shall be set to the null value of 0 since no talk group is associated with a unit to unit call. The remaining fields in the Header Word (MI, ALGID and KID) shall be set as defined in [1] and [5].

While transmitting on the inbound traffic channel, the SU shall fill the LC field with the LC_UU_V_CH_USR (Unit to Unit Voice Channel User) message as defined in [2]. The entire Service Options field as defined [2] shall be considered Reserved as defined in [4]. This means there is no conventional Emergency unit to unit call. As defined in [5] the LC field for unit to unit voice transmissions shall utilize a source unit ID and destination unit ID both in the range of 1 to 9,999,999.

As previously defined in 6.1.1.1 above the contents of Link Control and LSD fields may also be encrypted if the voice information is encrypted.

The fields in the ESYNC Word (MI, ALGID and KID) shall be set as defined in [1] and [5].

6.4.1.2 FNE Response

The following actions are valid responses from the FNE to a unit-to-unit voice transmission, though an FNE need not perform all of these actions:

- (Repeat Mode Only) The FNE may transparently repeat the unit-to-unit voice transmission as defined in section 3 above. The FNE may also take into account console wireline priority when determining whether to repeat the transmission.

While transmitting on the outbound channel in repeat mode, the FNE shall send the first LC_UU_V_CH_USR message received and then repeat subsequent LC_UU_V_CH_USR messages at least once every NTrafficLC LC messages. The source address shall reflect the currently active source address and the destination address shall reflect the currently active destination address.

6.4.1.3 Receiving Subscriber Unit Actions

Depending on its squelch setting, a subscriber monitoring the channel may display received call information and play out the audio associated with the received transmission. The following text explains how an SU determines if the received unit to unit call should be presented to the user.

The selective squelch mode along with the receiving subscriber's unit ID can be used to filter undesired unit to unit calls from the user.

The operation of the subscriber receiver for unit to unit voice messages consists of three main cases, with variations that depend on the receiver's operation. The three main cases are called Squelch conditions in this document. They are Monitor, Normal Squelch, and Selective Squelch. As in the case of the transmitter, the receiver operation is affected by the current settings of the subscriber as previously described in 6.1.1.3 above except for the replacement of the Talk Group(s) with the Receiving SU's Unit ID for unit-to-unit transmission.

The Monitor switch and the three types of squelch operation were previously defined in 6.1.1.3 above and also apply to unit to unit calls. The discussion about how a receiver should operate based on a specific set of attributes also applies to unit to unit calls.

The details of the procedure in the receiver for implementing these different squelch types are not covered here for the same reasons as previously defined in 6.1.1.3 above. The conditions for unmuting for a unit to unit voice transmission are given in Table 4 below for the three squelch settings. Note that a receiver may operate as if it is in Monitor mode (and not support the Normal or Selective squelch mode) when configured with a receive NAC of \$F7E.

Table 4 – Receiver Unmute Conditions for Unit to Unit Voice Transmissions

Attribute	Monitor	Normal Squelch	Selective Squelch
NAC	Don't Care	= Receiver NAC or Don't Care if Receiver NAC is \$F7E	= Receiver NAC or Don't Care if Receiver NAC is \$F7E
Destination ID	Don't Care	Don't Care	= Receiver ID
Encryption	Don't Care	Decryption is Possible	Decryption is Possible

The output of the receiver in the monitor mode when decryption is impossible operates as defined in 6.1.1.3 above.

The received Source ID may be displayed for use by the radio operator. When the source ID is displayed numerically, it shall be displayed with decimal digits. Instead of a numeric ID, the SU may display a textual phrase which has been mapped to the numeric source ID.

6.4.1.4 Call Termination

The transmitting subscriber shall terminate its voice transmission by sending a simple Terminator Data Unit (TDU) as defined in [4].

6.4.1.4.1 FNE Response

The FNE response is the same as defined in 6.4.1.4.1 above.

6.4.1.4.2 Receiving SU Actions

The receiving SU's actions are the same as defined in 6.1.1.4.2 above.

6.4.2 FNE Sourced Unit-to-Unit Voice Transmission

This is an FNE sourced unit-to-unit voice transmission, typically at the request of one of the FNE's consoles. The procedures are the same as 6.4.1 above with the originating FNE taking on the role of the originating SU. Note also that the source FNE may also adhere to the procedures described in 6.4.1.4.1 above.

6.5 Unit to PSTN Call Procedure

There are a number of alternative requests for a PSTN service, depending on whether the call is explicit or implicit dialing, and in the case of explicit dialing whether up to 10 or more digits are dialed. Explicit dialing is where the required number is dialed in full, while implicit dialing is where a short form code is dialed, which the FNE translates (for each subscriber) into a telephone number.

The subscriber performs the dialing operation as described in [11].

A typical call sequence diagram for unit to PSTN voice call is shown in Figure 14 below.

6.5.1 Unit to PSTN Service Request

An SU makes a call by generating a TELE_INT_DIAL_REQ or TELE_INT_PSTN_REQ on the conventional channel, complying with the channel access rules of 4.2.1 above. The SU may retry the message awaiting a response from the FNE; however the FNE may never respond. The following procedures utilize the telephone interconnect messages defined in [3].

6.5.1.1 FNE Response

The following messages may be generated by an FNE as valid call progress responses to a TELE_INT_DIAL_REQ or TELE_INT_PSTN_REQ. These messages are sent by the FNE using the channel access rules defined in 4.2.3 above. An SU need not support any of these messages:

- A QUE_RSP if the FNE cannot proceed with call establishment due to temporary lack of resources.
- A DENY_RSP if the FNE determines that the call cannot be permitted.
- An ACK_RSP_FNE if the FNE wishes to indicate it has received the request and is in the process of processing it.

It is recommended that the FNE also relay the call progress tones received from the telephone network to the user equipment.

6.5.1.2 Subscriber Unit Actions

An SU which is capable of receiving the above messages shall assume that the telephone interconnect request was successful if no messages are received in response to its request. SU's which are not capable of receiving the above messages shall also assume that the telephone interconnect request was successful.

If an SU attempting access or waiting for signaling receives a valid Unit to PSTN response then it shall take action as follows:

- If a QUE_RSP is received, the SU shall wait for further signaling, and may indicate to the user the progress of the call. A timer should exist in the SU to enable exiting this wait state.
- If a DENY_RSP is received, the SU shall return to the idle state, and may indicate to the user the reason for the failure of the call.
- If an ACK_RSP_FNE is received, the SU shall cease its retries and await further FNE responses. A timer should exist in the SU to enable exiting this wait state.

The Unit to PSTN response is considered valid if the source address of the message matches the SU's UID.

The SU plays out the audio it receives to the user as described in 2.2.4 of [11].

6.5.2 Traffic Channel Maintenance

While transmitting on the conventional channel, the SU shall fill the LC field with the LC_TELE_INT_V_CH_USR message. While transmitting on the outbound traffic channel, the FNE shall send the LC_TELE_INT_V_CH_USR message at least once every **NTrafficLC** LC messages.

Additional digits may be sent over the traffic channel while the telephone interconnect call is in progress. Refer to 2.4.3 in [11] for informative details.

6.5.3 Call Termination

The SU, the telephone network, or the FNE may terminate the call.

6.5.3.1 Subscriber Unit Actions

If the SU receives a call tear down message (ETDU/CTC per 7.3.8 of [2]) targeted to its UID the SU shall terminate the call.

An in progress interconnect call may be terminated when the user causes their SU to generate a pre-programmed disconnect code. Pre-programmed disconnect codes shall use one or more Telephone Interconnect Request - Explicit Dialing messages [3] matching a preprogrammed code. This preprogrammed code shall not be the same code as is used for initiating a hookflash. See [11] for additional information about hookflash.

An in progress interconnect call may be terminated when the user causes their SU to generate a CAN_SERV_REQ. To ensure interoperability, subscriber equipment capable of generating a CAN_SERV_REQ shall also be able to terminate a call using the disconnect code described in the paragraph above.

6.5.3.2 FNE Actions

A CAN_SERV_REQ received by a conventional FNE from an SU may cause disconnection of a call in progress.

A conventional FNE shall be capable of terminating a call in response to the dialing of a pre-programmed disconnect code as described in 6.5.3.1 above.

When a FNE disconnects a telephone call it shall cause a call tear down termination message (Expanded TDU) as defined in 7.3.8 of [2] to be sent on the conventional channel. An audible confirmation may be transmitted prior to the call tear down.

6.6 PSTN to Group Call Procedure

A typical call sequence diagram for PSTN to group voice call is shown in Figure 15 below. A call from the telephone network is received by the FNE. The means by which the FNE determines the call is a group call is not part of this specification.

How the FNE determines that the phone line has stopped transmitting and that it should end the PSTN transmission but leave it off hook so that the SU can talk back to the phone line is beyond the scope of this document.

6.6.1 FNE Action

The FNE shall provide an audible alert to the presence of someone accessing the FNE. The audible alert is heard by all users of the channel. The FNE then proceeds to start the call by generating outbound audio.

6.6.2 SU Response

The SU can respond by dialing a preprogrammed “accept code” or by simply making a group voice transmission on the inbound traffic channel as described in 6.1.1 above.

6.6.3 Conventional Channel Maintenance

While transmitting on the inbound traffic channel, the SU shall fill the LC field with the LC_GRP_V_CH_USR message as defined in [2]. While transmitting on the outbound traffic channel, the FNE shall send the LC_GRP_V_CH_USR message at least once every **NTrafficLC** LC messages. The source address shall reflect the currently active source address and the group address shall reflect the currently active group.

6.6.4 Call Termination

Only the telephone network or the FNE may terminate the call.

As a result of a call termination signal from the telephone network or an FNE timeout, the FNE shall clear down the call by transmitting one or more call termination (Expanded TDU as defined in [2] and terminating any further voice transmissions associated with this call.

6.7 PSTN to Unit Call Procedure

A typical call sequence diagram for PSTN to unit voice call is shown in Figure 16 below.

6.7.1 PSTN to Unit Service Request

A call from the telephone network is received by the FNE. The means by which the FNE determines the call is destined for an Individual on a particular channel is not part of this specification.

6.7.1.1 Availability Check on Called Party

Upon determining that a call from the telephone network to an individual is starting, the FNE may proceed without performing the Availability Check and connects the incoming phone call to the appropriate channel. This is defined in 2.3.3.2 and 3.7 of [11]. The remainder of this section covers the Availability Check procedure.

Upon receipt of a service request from the PSTN, the FNE may check the availability of the target unit by sending a TELE_INTER_ANS_REQ to the SU that is the target of the incoming phone call. The FNE should start a timer (recommended to be 1 second) and wait for the TELE_INT_ANS_RESP. While waiting for a response the FNE shall provide ring-back tone to the telephone network. If the timer expires and the target SU has not yet responded the call shall be ended and the FNE may within 1 second abandon the call attempt and connect the incoming phone call to a tone signal, recorded announcement, operator position or voice mail facility as may be appropriate.

If the SU responds with a DENY TELE_INTER_ANS_RSP the FNE shall end the call and connect the incoming phone call to a tone signal, recorded announcement, operator position or voice mail facility as appropriate to the telephone network it is connected to.

If the SU responds with a PROCEED TELE_INTER_ANS_RSP the FNE shall connect the incoming phone call to the conventional channel and if equipped to do so shall signal answer supervision to the telephone network.

An FNE that is capable of availability check procedures shall be capable of disabling these features for telephone calls to all or a subset of SUs.

6.7.2 Traffic Channel Maintenance

While transmitting on the conventional channel, the SU shall fill the LC field with the LC_TELE_INT_V_CH_USR message. While transmitting on the outbound traffic channel, the FNE shall send the LC_TELE_INT_V_CH_USR message at least once every NTrafficLC messages.

6.7.3 Call Termination

The telephone network, the called party, or the FNE may terminate the call.

6.7.3.1 Subscriber Unit Actions

If the SU receives a call tear down message (ETDU/CTC per 7.3.8 of [2]) targeted to its UID the SU shall terminate the call.

An in progress interconnect call may be terminated when the user causes their SU to generate a pre-programmed disconnect code as described in 6.5.3.1 above.

An in progress interconnect call may be terminated when the user causes their SU to generate a CAN_SERV_REQ. To ensure interoperability, subscriber equipment capable of generating a CAN_SERV_REQ shall also be able to terminate a call using the disconnect code described in 6.5.3.1 above.

6.7.3.2 FNE Actions

A CAN_SERV_REQ received by a conventional FNE from an SU may cause disconnection of a call in progress.

A conventional FNE shall be capable of terminating a call in response to the dialing of a pre-programmed disconnect code as described in 6.5.3.1 above.

When a FNE disconnects a telephone call it shall cause a call tear down termination message (Expanded TDU) as defined in 7.3.8 of [2] to be sent on the conventional channel. An audible confirmation may be transmitted prior to the call tear down.

Limited Use Only

7 Conventional Packet Data Procedures

This section specifies the procedures for data packet and response packet transmission over a conventional radio channel operating in the Direct Data, Repeated Data, or Conventional FNE Data configurations specified in [7]. For the remainder of this section, the term packet refers to either a data packet or response packet unless specifically stated otherwise and the acronym FS means Frame Sync whereas the FS definition in 1.3 above applies to all other sections of this document.

7.1 Channel Access for Packet Data Transmission

The transmission of a packet follows a polite procedure which minimizes collisions on the channel. This procedure waits until the channel is available before transmitting the packet. The channel is available if either of two conditions is true.

1. **Idle Status Symbol** is detected. In this case, the SU is listening to an outbound channel that has signaled that the inbound channel is idle. Idle Status Symbols are never sent in the Direct Data configuration as specified in 2.6 above.
2. **No channel activity** is detected. In this case, the SU is unable to hear any activity on the channel. This works in all packet data configurations.

The process that the SU uses to determine the inbound channel status and to access the channel in Repeated Data and Conventional FNE Data configurations is illustrated by the state diagram in Figure 12 below. The SU should always assume that the inbound channel state is Unknown until it has determined otherwise.

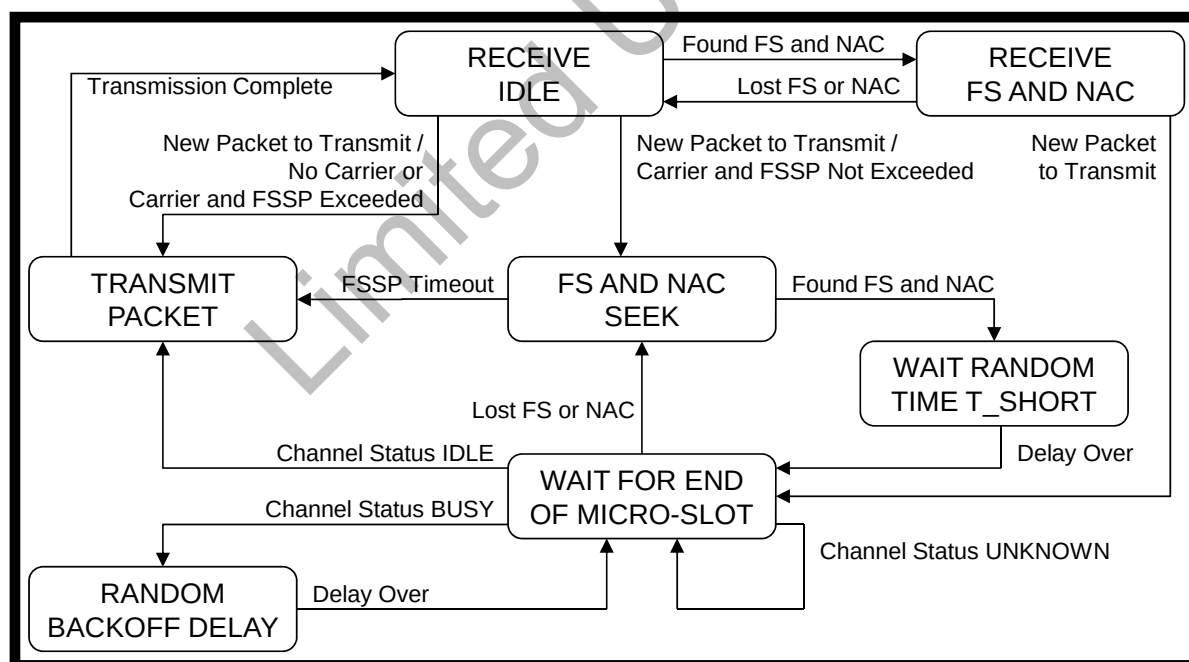


Figure 12– Conventional Data Channel Access State Diagram

SU transmit procedures are implemented to organize inbound transmissions in a manner which expedites the completion of voice or data communications transactions in progress and reduces the potential for collisions between contending SUs. These procedures are described below, with reference to Figure 12 above.

Generally, an idle SU oscillates between the RECEIVE IDLE and RECEIVE FS AND NAC states until it has a packet to transmit.

7.1.1 RECEIVE IDLE State

While the SU is in this state and it has nothing to transmit, it shall respond to the arrival of a Frame Sync pattern by decoding the NID. If there is a NAC match, the SU shall go to the RECEIVE FS AND NAC state.

If the SU has a new packet to transmit and it has not detected carrier, the SU shall go to the TRANSMIT PACKET state.

If the SU has a new packet to transmit and it has detected carrier for at least FSSP, the SU shall go to the TRANSMIT PACKET state.

If the SU has a new packet to transmit and it has detected carrier for less than FSSP, the SU shall go to the FS AND NAC SEEK state.

7.1.2 RECEIVE FS AND NAC State

While in this state, the SU shall decode the NID and header block of each packet in order to recognize the NAC and its own ID. If appropriately addressed packets occur they are received.

If there are no further Frame Sync patterns or the NAC does not match then the SU shall go to the RECEIVE IDLE state.

If the SU has a new packet to transmit, the SU shall go to the WAIT FOR END OF MICRO-SLOT state.

7.1.3 FS AND NAC SEEK State

While in this state, the SU tries to locate a Frame Sync sequence on the outbound channel so that it can determine the location of the channel status symbols.

The SU shall respond to the arrival of a Frame Sync sequence by decoding the NID. If there is a NAC match, the SU shall go to the WAIT RANDOM TIME T_SHORT state.

If no Frame Sync and NAC match is detected for the Frame Sync Seek Period (FSSP) time limit, then the SU shall go to the TRANSMIT PACKET state and begin transmission.

7.1.4 WAIT RANDOM TIME T_SHORT State

Upon entering this state, the SU waits a random delay time, uniformly distributed between 0 and T_SHORT. The SU then goes to the WAIT FOR END OF MICRO-SLOT state.

This delay serves to reduce the collision potential with other SUs that may be exiting the FS AND NAC SEEK state at the same time.

7.1.5 WAIT FOR END OF MICRO-SLOT State

Upon entering this state the SU shall delay until the end of the current micro-slot. The channel status symbol at this boundary is either IDLE, BUSY, or UNKNOWN.

If it is IDLE then the SU may go to the TRANSMIT PACKET state and begin transmission.

If it is BUSY then the SU shall enter the RANDOM BACKOFF DELAY state.

If it is UNKNOWN then the SU shall return to this state.

If the end of the outbound transmission occurs and the SU loses sync with the outbound signal for some reason, then it shall go to the FS AND NAC SEEK state.

7.1.6 RANDOM BACKOFF DELAY State

Upon entering this state, the SU shall delay a random time uniformly distributed between 0 and T_LONG for data packets, or 0 and T_ACK for response packets. After the delay is over the SU returns to the WAIT FOR END OF MICRO-SLOT state.

The random delay redistributes the subsequent accesses of multiple contending SUs into the future to reduce collision potential and prevent them from accessing the channel in lock-step. Note that the maximum delay is shorter for response packets than for data packets, which gives them priority access to the inbound channel.

7.1.7 TRANSMIT PACKET State

Upon entering this state, the SU shall transmit the packet and then go to the RECEIVE IDLE state.

7.1.8 Channel Access Parameters

Table 5 below lists the parameters for the channel access procedure. These are recommended default values. Different systems may be tuned to operate with different values to improve performance.

The FSSP time limit should be at least as long as the longest possible time interval between Frame Sync patterns on the channel. The FSSP time out value in Table 5 below is equal to the transmit time for a confirmed delivery data packet with 512 octets of data. Some systems may choose to shorten or lengthen this time if the system applications impose different constraints on packet sizes. Systems with unconfirmed data packets with up to 512 octets of data should use FSSP = 938 ms since unconfirmed data packets use a lower coding rate.

Table 5 – Recommended Channel Access Procedure Parameters

Name	Description	Value
FSSP	Frame Sync Seek Period	728 ms
Channel Access Delay spreads:		
T_ACK	- for responses after BUSY	250 ms
T_LONG	- for all data packets after BUSY	500 ms
T_SHORT	- after first frame sync	50 ms

7.2 Receiver NAC Qualification

The NAC is allowed if it equals the receiver's NAC or if the receiver is configured with a receive NAC of \$F7E. This qualification is performed as part of Receiver Preprocessing as specified in [8].

7.3 Packet Data Conveyance

Packets shall be conveyed over conventional radio channels in accordance with [7].

8 Supplementary Data Procedures

The procedures for these messages generally follow the trunking procedures described in [10]. There are some differences for conventional LMR systems, which abbreviate the given trunking procedures.

Because there is no control channel, the random access protocol for the trunking control channel is not followed. Instead, the channel access procedure given in 4.2 above shall be followed. These procedures are intended to follow a polite channel access procedure. In some cases, such as Emergency Alarm, a combined impolite and polite channel access procedure may be followed instead. The decision is up to the manufacturer.

8.1 Emergency Alarm Procedures

A typical Emergency Alarm sequence diagram is shown in Figure 17 below.

8.1.1 Overview

Emergency declarations can be initiated in several ways as detailed below.

An emergency is a heightened state of alert for an SU. Emergency conditions are separated into two types:

- Emergency Alarm – A notification from an SU to a dispatch console (via the FNE) indicating that the user of this SU has an emergency condition.
- Emergency Call – A voice transmission processed in emergency mode.

SUs initiate Emergency Alarm and Emergency Call. A Console may initiate an Emergency Call. Consoles should not initiate Emergency Alarms.

The SU is responsible for notifying the FNE of an emergency condition and the FNE is responsible for acknowledging receipt of the notification. The FNE is additionally responsible for notifying consoles and other wireline monitoring equipment of new emergency declarations, and may maintain the state of units/groups with regard to emergency status. The FNE may cancel its internal emergency condition of an SU based on receiving a cancellation request from a dispatch console, but the FNE cannot cancel the internal emergency condition of an SU (see below).

The SU is responsible for maintaining its own emergency status, and monitoring and may display in-progress emergencies of other SUs. An in-progress emergency is defined as an emergency notification from an SU that has been received by a different SU. A conventional SU does not know when a different SU has canceled its own emergency. Depending on the manufacturer's implementation, a conventional console can be aware of other consoles canceling the emergency state of the FNE with respect to a particular SU.

The SU notifies the FNE of an emergency condition through the following mechanisms:

- The Emergency Alarm message and procedure
- Group call procedures using the Emergency status bit (b7) in the Service Options field

There are three combinations of the above mechanisms. An SU may support one or more of these mechanisms, the one used in operation is chosen through the SUs configuration.

- **Emergency Call Mode:** A user can put their SU in emergency mode and then all subsequent group voice transmissions are sent with the Emergency status bit (b7) in the Service Options field set.
- **Emergency Alarm Mode:** A user can put their SU in emergency mode which causes it to send an EMRG_ALARM_REQ and all subsequent voice transmissions are sent with the Emergency status bit (b7) in the Service Options field cleared (non emergency mode).
- **Emergency Alarm and Call Mode:** A user can put their SU in emergency mode which causes an EMRG_ALARM_REQ to be sent and all subsequent voice transmissions are sent with the Emergency status bit (b7) in the Service Options field set (emergency mode).

Dispatch consoles are also typically involved in emergency acknowledgements, cancellation, and display of emergency conditions. When the console receives an emergency notification it may emit audible and/or visual indications. A dispatch operator acknowledges the emergency situation (method not defined in this specification and is manufacturer dependent) which usually includes stopping the audible and/or visual indications (i.e. human acknowledgement). The fact that the FNE has received the notification of the emergency condition is indicated to the initiating SU by various means described in the following sections. The fact that a dispatch operator has also acknowledged receipt may trigger an indication back to the initiator but this is outside the scope of the standard. At this point the emergency is still considered in-progress. Additional human actions are expected to terminate the emergency condition as described in the detailed procedures below.

In the case of a group Emergency Call voice transmission, the emergency bit is set based on the device configuration settings and the user having previously set their device into emergency mode. Each device is responsible for setting or clearing their emergency setting. Thus while some voice transmissions on a channel may have the emergency bit set, others on the same channel may not. Further detail regarding Emergency Call can be found in 6.2 above.

8.1.2 Emergency Alarm Request

An SU shall make an Emergency Alarm request by generating an EMRG_ALARM_REQ on the channel complying with the channel access protocol defined in 4.2.2 above.

The FNE takes the following action:

- The FNE shall acknowledge the Alarm Request by sending an ACK_RSP_FNE to the SU and may send notification of a new emergency declaration by this SU to consoles and other wireline monitoring equipment. This notification may include the group identified in the EMRG_ALARM_REQ. The ACK_RSP_FNE to the SU may be based on FNE confirmation of delivery of the emergency alarm request message to a dispatch console.

- (Repeat Mode only) The FNE may transparently repeat the EMRG_ALARM_REQ as defined in section 3 above. The FNE may also take into account console wireline priority when determining whether to repeat the EMRG_ALARM_REQ.
- The FNE may begin to also track the emergency condition.

The originating SU takes the following action:

- The SU may impolitely retry the EMER_ALARM_REQ message up to NIRetry times followed by NPRetry polite retries or until the appropriate ACK_RSP_FNE acknowledgement as defined in [3] is received by the SU.
- The SU may notify the user of success or failure of the Emergency Alarm request.

The SU may activate an internal emergency alarm condition. Subsequent group call requests made by this SU while the internal emergency alarm condition is active may be made in emergency mode (i.e. set the “E” bit in the Link Control field).

8.1.3 Emergency Alarm Cancellation

An SU may cancel its own Emergency Alarm and inform the FNE. The SU typically requires a special user initiated procedure to cancel an emergency. Details of the procedure whereby the user initiates an emergency alarm cancel are not within the scope of this document. When the SU Emergency Alarm cancel is executed the SU and FNE may perform the following procedure.

- The SU attempts to send a CAN_SRV_REQ on the inbound channel, complying with the channel access protocol in 4.2.1 above and containing:
 - o AIV=1 (Additional information field valid)
 - o Service Type = Emergency Alarm Request Opcode = %100111
 - o Reason Code set to \$00 (no reason)
 - o Additional Information set to: octet 4 = %1xxx xxxx echoing the original alarm request with emergency bit set to “true,” and octets 5-6 set to the 16-bit group address
 - o Source Address = SU Unit ID

If a CAN_SRV_REQ is received by the FNE with the information shown above, then:

- If the SU is valid on this FNE, the FNE shall respond with an ACK_RSP_FNE to the SU. The FNE may send notification of the emergency condition cancellation for this SU to consoles and other wireline monitoring equipment. This notification may include the group identified in the CAN_SRV_REQ.

8.2 Call Alert Procedures

Call Alert is a non-voice operation in which an originator can indicate its identity to the SU of another user with an indication to contact the indicated originator. This is typically used if the user of the target SU did not respond to a call from the originator. At a later time, the alerted (target) party may then return the call to the alerting (originating) party. Call Alerts can be sent from SU to SU which includes: mobile radio to mobile radio, mobile radio to console or console to mobile radio.

A typical Call Alert call sequence diagram is shown in Figure 19 below and Figure 20 below.

8.2.1 Initial Call Alert Sourced by a Mobile SU

8.2.1.1 Mobile Subscriber Unit Actions

The user shall select the SU to which to direct a Call Alert operation. (The description of the actual mechanism to select the Call Alert target address is beyond the scope of the standard and is left to the manufacturer).

The SU shall generate a CALL_ALERT_REQ complying with the channel access protocol of 4.2.1 above. The CALL_ALERT_REQ shall indicate the SU address for both the target SU and the initiating SU.

8.2.1.2 FNE Response

The following actions are valid responses from the FNE to a CALL_ALERT_REQ, though an FNE need not perform all of these actions:

- (Repeat Mode Only) The FNE may transparently repeat the CALL_ALERT_REQ as defined in section 3 above. The FNE may also take into account console wireline priority when determining whether to repeat the CALL_ALERT_REQ.

8.2.1.3 Initiating Mobile Subscriber Unit Actions

If an SU attempting access or waiting for signaling receives an appropriate response, then it shall take action as follows:

- If an ACK_RSP_FNE for the Call Alert operation is received indicating acceptance of the service request by the target unit, the SU shall return to the idle state, and may indicate the success to the user.
- The SU shall politely re-transmit the CALL_ALERT_REQ message until transmitting the maximum number of polite retries, NPRetry, or a until a valid response is received from the target SU. If it has exhausted its retries the originating SU shall return to the idle condition, and may indicate to the user the reason for the failure of the operation.

8.2.1.4 Target Subscriber Unit Actions

The target SU can be either a mobile SU or an FNE device such as a console. An SU receiving a CALL_ALERT_REQ message with target address matching the UID of the SU:

- may return an ACK_RSP_FNE using the channel access procedures defined in 4.2.3 above, addressed to the initiating SU as the Target, and itself as the Source, and display an indication of the Call Alert condition for the SU user. Note that the FNE may originate the ack as well if the destination was a console or a device assigned to the General FNE Address.
- may ignore the message if the SU does not support the service.

8.2.1.5 FNE Response to ACK_RSP_FNE Sourced by an SU

The following actions are valid responses to an ACK_RSP_FNE, though an FNE need not perform all of these actions:

- (Repeat Mode Only) An FNE may transparently repeat the ACK_RSP_FNE as defined in section 3 above. The FNE may also take into account console wireline priority when determining whether to repeat the ACK_RSP_FNE

8.2.2 Initial Call Alert Sourced by the FNE

8.2.2.1 FNE Actions

When directed by a console the FNE shall transmit the CALL_ALRT message indicating the SU address for both the target SU and initiating SU, following the channel access procedures of 4.3.1 above. The FNE may record that the target SU has a pending Call Alert.

Independently, or at the request of the console, the FNE shall repeat the CALL_ALRT message until a maximum number of polite retries, NPRetry, or a valid response is received from the target SU.

If there is no response, the FNE and/or console may consider the Call Alert to have failed and may indicate this to the requesting User.

If an ACK_RSP_FNE message is received from the target SU to the requesting SU, the FNE shall cancel the Call Alert process and return to normal processing for this addressed SU. The FNE / Console may indicate that the Call Alert process was successful for the addressed SU.

8.2.2.2 Target Subscriber Unit Actions

An SU receiving a CALL_ALRT message with target address matching the UID of the SU:

- may return an ACK_RSP_FNE using the channel access procedures defined in 4.2.3 above, addressed to the initiating SU as the Target, and itself as the Source, and display an indication of the Call Alert condition for the SU user; or
- may ignore the message if the SU does not support the service.

8.2.2.3 FNE Response to ACK_RSP_FNE Sourced by an SU

The following actions are valid responses to an ACK_RSP_FNE, though an FNE need not perform all of these actions:

- (Repeat Mode Only) An FNE may transparently repeat the ACK_RSP_FNE as defined in section 3 above. The FNE may also take into account console wireline priority when determining whether to repeat the ACK_RSP_FNE.

8.3 Radio Check Procedures

Radio Check can be used to determine if an addressed SU is currently active and/or available to the FNE. A response to the request for radio check is required by the FNE, otherwise the FNE shall consider the addressed SU to not be available. An SU may send a Radio Check to another SU. An SU shall not send a Radio Check to the General FNE address.

A typical Radio Check call sequence diagram is shown in Figure 18 below.

8.3.1 Initial Radio Check Sourced by the FNE

8.3.1.1 FNE Actions

When directed, the FNE shall use polite channel access methods as defined in 4.3.1 above to transmit the EXT_FNCT_CMD message associated with Radio Check and address this to the UID of the SU to be checked. The FNE may record that the SU is currently under check.

The FNE shall re-transmit the Radio Check EXT_FNCT_CMD message until a maximum number of retries, NPRetry, or a valid response is received from the target SU.

If there is no response, the FNE may consider the addressed SU to not be present on the FNE and may remove the SU, and all its influence upon activity at this FNE.

If a Radio Check ACK EXT_FNCT_CMD message is received indicating a valid response from the addressed unit, the FNE shall cancel the radio check process and return to normal processing for this addressed SU. The FNE may indicate that the Radio Check process was successful for the addressed SU.

The FNE may also transparently repeat the Radio Check ACK EXT_FNCT_CMD as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.

8.3.1.2 Target Subscriber Unit Actions

The SU may ignore the Radio Check EXT_FNCT_CMD message if it does not support the feature. The rest of the text describes the actions of an SU that supports Radio Check.

When an SU receives a Radio Check EXT_FNCT_CMD message addressed to the UID of this SU, the SU shall generate a valid response to the FNE.

If the SU is currently idle, the valid response shall be the Radio Check ACK EXT_FNCT_CMD message. This shall be sent in accordance to the access scheme described 4.2.3 above. The SU shall send this message only once and then shall resume normal idle processing.

If the SU is directed to initiate a service request while in the radio check process, the SU shall resume normal processing and send the service request message (in accordance to the access scheme described in 4.1 above).

8.3.2 Initial Radio Check Sourced by a Mobile SU

8.3.2.1 Initiating Subscriber Actions

The SU shall use polite channel access methods as defined in 4.2.1 above to transmit the EXT_FNCT_CMD message associated with Radio Check and address this to the UID of the SU to be checked. The SU may record that the SU is currently under check.

The SU shall re-transmit the Radio Check EXT_FNCT_CMD message until a maximum number of retries, NPRetry, or a valid response is received from the target SU.

If there is no response, the SU may consider the addressed SU to not be present in its coverage area.

If a Radio Check ACK EXT_FNCT_CMD message is received indicating a valid response from the addressed unit, the SU shall cancel the radio check process and return to normal processing for this addressed SU. The SU may indicate that the Radio Check process was successful for the addressed SU.

8.3.2.2 Receiving FNE Actions

The FNE may transparently repeat the Radio Check ACK EXT_FNCT_CMD as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.

8.3.2.3 Target Subscriber Unit Actions

The SU may ignore the Radio Check EXT_FNCT_CMD message if it does not support the feature. The rest of the text describes the actions of an SU that supports Radio Check.

When an SU receives a Radio Check EXT_FNCT_CMD message addressed to the UID of this SU, the SU shall generate a valid response to the FNE.

If the SU is currently idle, the valid response shall be the Radio Check ACK EXT_FNCT_CMD message. This shall be sent in accordance to the access scheme described 4.2.3 above. The SU shall send this message only once and then shall resume normal idle processing.

If the SU is directed to initiate a service request while in the radio check process, the SU shall resume normal processing and send the service request message (in accordance to the access scheme described in 4.1 above).

8.4 Radio Inhibit Procedures

Radio Inhibit can be used to cancel User service processing at the addressed SU. Essentially an "inhibited" SU shall appear to a user as if it were powered off. The SU shall not present any feedback to the user interface, nor accept any input from the user interface. The addressed SU shall respond to the Radio Inhibit command, but the FNE may proceed in the inhibited condition for this addressed SU with or without the explicit acknowledge response from the addressed SU. The inhibited SU shall not participate in any service activity to which it would normally be party. Subscriber units should not send the Radio Inhibit EXT_FNCT_CMD message.

A typical Radio Inhibit call sequence diagram is shown in Figure 18 below.

8.4.1 FNE Actions

When directed, the FNE shall adhere to the channel access rules defined in 4.3.1 above and transmit the EXT_FNCT_CMD message associated with Radio Inhibit and address this to the UID of the SU to be inhibited. The FNE may record that the SU has a pending Radio Inhibit.

The FNE shall re-transmit the Radio Inhibit EXT_FNCT_CMD message until a maximum number of polite retries (see 4.3.1 above), NPRetry, or an appropriate response is received from the target SU.

If there is no response, the FNE may consider the addressed SU to be inhibited and retain a pending Radio Inhibit condition for this SU, such that the next time the SU makes its presence known (e.g., the unit transmits) the FNE may start the Radio Inhibit message again.

If a Radio Inhibit ACK EXT_FNCT_CMD message is received, the FNE shall cancel the radio inhibit process and consider this addressed SU to be inhibited. The FNE may indicate that the Radio Inhibit process was successful for the addressed SU. (Repeat Mode Only) An FNE may transparently repeat the Radio Inhibit ACK EXT_FNCT_CMD as defined in section 3 above. The FNE may also take into account console wireline priority when determining whether to repeat the message.

The FNE may ignore any service request from an inhibited SU with the exception of:

- Extended Function Command responses

These transactions shall be processed normally, but shall not alter the current inhibited condition of the SU.

8.4.2 Target Subscriber Unit Actions

The SU may ignore the Radio Inhibit EXT_FNCT_CMD message if it does not support the feature. The rest of the text describes the actions of an SU that supports Radio Inhibit.

When an SU receives a Radio Inhibit EXT_FNCT_CMD message addressed to the current UID of this SU, the SU shall:

- Respond with the Radio Inhibit ACK EXT_FNCT_CMD message sent in accordance to the access scheme described in 4.2.3 above and,
- Assume the inhibited condition.

In the inhibited condition, the SU shall blank any User display device(s). There shall be no indication, visual or audible, of operational conditions of the SU to the User. The SU shall ignore all User or other external input.

An inhibited SU shall ignore all messages with the exception of:

- Extended Functions Command. The SU shall act according to the extended function value, e.g. it shall return the Radio Check Acknowledge extended function message in response to a Radio Check Extended Function Command.
- OTAR commands

An inhibited SU shall retain the inhibited condition through power cycling. This condition shall only be altered by reception of a valid Radio Uninhibit Extended Function Command, or explicit field reprogramming of the SU.

8.5 Radio Uninhibit Procedures

Radio Uninhibit is sent to cancel a prior Radio Inhibit condition at the addressed SU. The FNE shall re-establish the addressed SU as a normal unit, capable of service requesting again. The addressed SU is expected to respond to the Radio Uninhibit message. The addressed SU shall now indicate normal operational conditions to the User, accept service requests from the User, and participate in normal activity. Subscriber Units should not send the Radio Uninhibit EXT_FNCT_CMD message.

A typical Radio Uninhibit call sequence diagram is shown in Figure 18 below.

8.5.1 FNE Actions

When directed, the FNE shall adhere to the channel access rules defined in 4.3.1 above and transmit the EXT_FNCT_CMD message associated with Radio Uninhibit and address this to the UID of the SU to be uninhibited. The FNE may record that the SU is has a pending Radio Uninhibit.

The FNE shall retransmit the Radio Uninhibit EXT_FNCT_CMD message until a maximum number of polite retries (see 4.3.1 above) NPRetry, or a valid response is received from the target SU.

If a valid response Radio Uninhibit ACK EXT_FNCT_CMD message is received, the FNE shall indicate that the SU is not inhibited. The FNE may also transparently repeat the Radio Uninhibit ACK EXT_FNCT_CMD as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.

8.5.2 Target Subscriber Unit Actions

The SU may ignore the Radio Uninhibit EXT_FNCT_CMD message if it does not support the feature. The rest of the text describes the actions of an SU that supports Radio Uninhibit.

When an SU receives a Radio Uninhibit EXT_FNCT_CMD message addressed to the UID of the SU, this SU shall:

- Respond with Radio Uninhibit ACK EXT_FNCT_CMD message in accordance to the channel access scheme described in 4.2.3 above.
- Cancel any existing inhibit condition, returning to normal operational presentation to the User; and
- Resume normal idle processing

If the SU is not inhibited and receives an uninhibit order, it shall still acknowledge the command and remain at current state.

8.6 Message Update Procedures

An SU or console can leave short messages to another subscriber by using the Message Update procedure. The SU may either display the message codes or the messages directly by finding the message in a code table using the message code and the identity of the sender.

A typical Message Update call sequence diagram is shown in Figure 21 below, Figure 22 below, and Figure 23 below.

8.6.1 Initial Message Update Sourced by a Mobile SU

8.6.1.1 Mobile Subscriber Unit Actions

An SU shall send a message code by generating a MSG_UPDT_REQ message to the FNE complying with the channel access procedures described in 4.2.1 above. The MSG_UPDT_REQ identifies the SU making the message update, the message being reported, and the target (unit, group or General FNE Address) to respond to this message update.

8.6.1.2 FNE Actions

The following actions are valid FNE responses to a MSG_UPDT_REQ, though a FNE need not perform all of these actions:

- The FNE may transparently repeat the MSG_UPDT_REQ as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.
- If destined for a talk group, the FNE shall acknowledge the MSG_UPDT_REQ by sending an ACK_RSP_FNE addressing the source SU of the MSG_UPDT_REQ. The sending of an acknowledgement may be based on delivery of the message to an interested console. Having the FNE acknowledge the group destined message reduces the number of devices simultaneously sending an acknowledgement.
- The FNE may originate the ack if the destination was a device assigned to the General FNE Address.

8.6.1.3 Initiating Mobile Subscriber Unit Actions

If an SU attempting access or waiting for signaling receives a valid response, then it shall take action as follows:

- If an ACK_RSP_FNE is received indicating acceptance of the service request by the target unit, the SU shall return to the idle state, and may indicate the success to the user.
- The SU shall repeat the MSG_UPDT_REQ message until a maximum number of retries, NPRetry, or a valid response is received from the target.
- If it has exhausted its retries the originating SU shall return to the idle condition, and may indicate to the user the reason for the failure of the operation.

8.6.1.4 Target Subscriber Unit Actions

The target SU can be either a mobile SU or an FNE device such as a console. The SU addressed by this message shall acknowledge reception of the message. If an SU detects a MSG_UPDT_REQ addressed to its current UID or its group ID, the SU:

- may store the message information from the initiating SU.
- may send an ACK_RSP_FNE according to the access rules defined in 4.2.3 above, addressing the source SU if the MSG_UPDT_REQ was destined for its Source ID
- may display the message information to the User.
- may ignore the message if the SU does not support this service.

If it supports the feature, the SU may retain the message information until it is cleared by the user.

8.6.1.5 FNE Response to ACK_RSP_FNE Sourced by an SU

The following actions are valid responses to an ACK_RSP_FNE, though a FNE need not perform all of these actions:

- (Repeat Mode Only) An FNE may transparently repeat the ACK_RSP_FNE as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.

8.6.2 Initial Message Update Sourced by the FNE

8.6.2.1 FNE Actions

When directed by a console the FNE shall comply with the channel access procedures described in 4.3.1 above and generate the MSG_UPDT message indicating the address of the target, the address of the initiating SU, and the indicated message number. The FNE shall record that the SU is has a pending Message Update.

The FNE shall retransmit the MSG_UPDT message until a maximum number of polite retries (see 4.3.1 above), NPRetry, or a valid response is received from the target SU.

If there is no response, the FNE and/or console may consider the Message Update to have failed and may indicate this to the requesting User.

If an ACK_RSP_FNE message is received from the target SU to the requesting console, the FNE shall cancel the Message Update process and return to normal processing for this addressed SU. The FNE may indicate that the Message Update process was successful for the addressed SU.

The destination ID of MSG_UPDT shall always be a 24 bit unit ID (UID) in the range of 1 to 9,999,999 when it is sourced by an FNE (at the request of a console).

8.6.2.2 Target Subscriber Unit Actions

The SU addressed by this message acknowledges reception of the message if it supports the feature. If an SU detects a MSG_UPDT addressed to its current UID, the SU:

- may display the message information to the user,
- may send an ACK_RSP_FNE that complies with 4.2.3 above, addressing the initiating SU of the MSG_UPDT as the Target, and itself as the Source.
- may ignore the message if it doesn't support the feature.

If it supports the feature, the SU shall retain the message information until it is cleared by the user.

8.6.2.3 FNE Response to ACK_RSP_FNE sourced by an SU

The following actions are valid responses to an ACK_RSP_FNE, though a FNE need not perform all of these actions:

- (Repeat Mode Only) An FNE may transparently repeat the ACK_RSP_FNE as defined in section 3 above. The FNE may also consider console wireline priority when deciding whether to repeat the message

8.7 Status Update Procedures

This describes the actions used to allow an SU to indicate its current status (User and Unit status) to a designated target address. Consoles do not originate Status Update messages; however an FNE may send one based on input from another FNE.

A typical Status Update call sequence diagram is shown in Figure 21 below, Figure 22 below, and Figure 23 below.

8.7.1 Initial Status Update Sourced by a Mobile SU

8.7.1.1 Mobile Subscriber Unit Actions

A mobile SU shall transmit a status update by generating a STS_UPDT_REQ complying with the channel access protocol of 4.2.1 above. The STS_UPDT_REQ identifies the SU making the status update, the status being reported, and the target (unit, group or General FNE Address) to respond to this status update.

8.7.1.2 FNE Actions

The following actions are valid responses to a STS_UPDT_REQ, though an FNE need not perform all of these actions:

- The FNE may transparently repeat the STS_UPDT_REQ as defined in section 3 above. The FNE may also take into account console wireline priority when deciding whether to repeat the message.
- If destined for a talk group, the FNE shall acknowledge the STS_UPDT_REQ by sending an ACK_RSP_FNE addressing the source SU of the STS_UPDT_REQ. The sending of an acknowledgement may be based on delivery of the message to an interested console. Having the FNE acknowledge the group destined message reduces the number of devices sending an acknowledgement simultaneously.
- The FNE may originate the ack if the destination was a device assigned to the General FNE Address.

8.7.1.3 Initiating Mobile Subscriber Unit Actions

If an SU attempting access or waiting for signaling receives a valid response, then it takes action as follows:

- If an ACK_RSP_FNE is received indicating acceptance of the service request by the target, the SU shall indicate the acknowledged status as the current status, return to the idle state, and may indicate the success to the user.
- The SU shall retransmit the STS_UPDT_REQ message until a maximum number of polite retries, NPRetry, or a valid response is received from the target. If it has exhausted its retries the originating SU shall return to the idle condition, and may indicate to the user the reason for the failure of the operation.

8.7.1.4 Target Subscriber Unit Actions

The target SU can be either a mobile SU or an FNE device such as a console. If an SU detects a STS_UPDT_REQ addressed to its current UID or GID, the SU:

- may store the status information from the addressed SU.
- may adhere to the channel access rules of 4.2.3 above and send an ACK_RSP_FNE addressing the source SU if the STS_UPDT_REQ was destined for its Source ID
- may display the status information to the User.
- may ignore the message if the device does not support this service.

The SU may retain the status for this SU until it is cleared by the User.

8.7.1.5 FNE Response to ACK_RSP_FNE Sourced by an SU

The following actions are valid responses to an ACK_RSP_FNE, though an FNE need not perform all of these actions:

- (Repeat Mode Only) An FNE may transparently repeat the ACK_RSP_FNE as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.

8.7.2 Initial Status Update Sourced by the FNE

8.7.2.1 FNE Actions

When caused by input from another FNE, an FNE shall comply with the channel access procedures described in 4.3.1 above and generate the STS_UPDT message indicating the address of the target, the SU address of the initiating SU, and the indicated status. The FNE may record that the SU has a pending Status Update.

The FNE shall repeat the STS_UPDT message until a maximum number of polite retries (see 4.3.1 above), NPRetry, or a valid response is received from the target SU.

If there is no response, the FNE may consider the Status Update to have failed and may indicate this to the requesting User.

If an ACK_RSP_FNE message is received from the target SU to the requesting SU, the FNE shall cancel the Status Update process and return to normal processing for this addressed SU. The FNE may indicate that the Status Update process was successful for the addressed SU.

8.7.2.2 Target Subscriber Unit Actions

The SU addressed by this message acknowledges reception of the message if it supports the feature. If an SU detects a STS_UPDT addressed to its current UID, the SU:

- may display the status information to the user.
- may send an ACK_RSP_FNE that complies with 4.2.3 above, addressing the initiating SU of the STS_UPDT as the Target, and itself as the Source.
- may ignore the message if it doesn't support the feature.

If it supports the feature, the SU may retain the status information until it is cleared by the user.

8.7.2.3 FNE Response to ACK_RSP_FNE Sourced by an SU

The following actions are valid responses to an ACK_RSP_FNE, though a FNE need not perform all of these actions:

- Repeat Mode Only) An FNE may transparently repeat the ACK_RSP_FNE as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.

8.8 Status Query Procedures

Status Query allows an SU to request the current status (User and Unit status) of a specified SU. Typically this message is sent from a console (FNE) to a particular mobile radio. However, the request can also be sent from one mobile radio to another mobile radio. Consoles cannot be queried for their "status".

A typical Status Query call sequence diagram is shown in Figure 24 below and Figure 25 below.

8.8.1 Initial Status Query Sourced by the FNE

8.8.1.1 FNE Actions

When directed by a console the FNE shall comply with the channel access procedures described in 4.3.1 above and generate the STS_Q message indicating the SU address for both the target SU and initiating SU. The FNE may record that the SU has a pending Status Query.

The FNE shall retransmit the STS_Q message until a maximum number of polite retries (see 4.3.1 above), NPRetry, or a valid response is received from the target SU.

If there is no response, the FNE and/or console may consider the Status Query to have failed and may indicate this to the requesting User.

If a STS_Q_RSP message is received from the target SU to the requesting SU, the FNE shall cancel the Status Query process and return to normal processing for this addressed SU. The FNE may indicate that the Status Query process was successful for the addressed SU.

8.8.1.2 Target Subscriber Unit Actions

An SU receiving the STS_Q message with target address matching the UID of the SU shall either:

- Adhere to the channel access rules of 4.2.3 above and return an STS_Q_RSP; or
- Ignore the message if the SU does not support this service.

8.8.1.3 FNE Response to STS_Q_RSP from an SU

The following actions are valid responses to a STS_Q_RSP, though an FNE need not perform all of these actions:

- (Repeat Mode Only) An FNE may transparently repeat the STS_Q_RSP as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.

8.8.2 Initial Status Query Sourced by a Mobile SU

8.8.2.1 Subscriber Unit Actions

An SU sends a status query request by generating a STS_Q_REQ complying with the channel access protocol of 4.2.1 above.

8.8.2.2 FNE Response to STS_Q_REQ Sourced by SU

The following actions are valid responses to a STS_Q_REQ, though an FNE need not perform all of these actions:

- The FNE may transparently repeat the STS_Q_REQ as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.

8.8.2.3 Initiating Subscriber Unit Operation

If an SU attempting access or waiting for signaling receives a valid response, then it shall take action as follows:

- If a STS_Q_RSP is received indicating acceptance of the service request by the target unit, the SU shall return to the idle state, may store the reported status for the addressed SU, and may indicate the success to the user.
- The SU shall retransmit the STS_Q_REQ message until a maximum number of polite retries, NPRetry, or a valid response is received from the target. If it has exhausted its retries the originating SU shall return to the idle condition, and may indicate to the user the reason for the failure of the operation.

8.8.2.4 Target Subscriber Unit Actions

An SU receiving the STS_Q_REQ message with target address matching the UID of the SU shall either:

- Adhere to the channel access rules of 4.2.3 above and return an STS_Q_RSP; or
- Ignore the message if the SU does not support this service.

8.8.2.5 FNE Response to STS_Q_RSP from an SU

The following actions are valid responses to a STS_Q_RSP, though an FNE need not perform all of these actions:

- (Repeat Mode Only) An FNE may transparently repeat the STS_Q_RSP as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.

8.9 Radio Unit Monitor Procedures

Radio Unit Monitor is an operation to cause a mobile subscriber radio to initiate a call when requested to do so by a dispatcher. The purpose of the service is to allow the dispatcher to listen to activity at the location of the SU. The originator of the procedure is the dispatcher and the target of the procedure is the SU. The dispatcher selects the target SU to which it wants to direct a radio unit monitor operation. The dispatcher may be resident in the FNE at a wireline dispatch point, or at an SU.

A typical Radio Unit Monitor call sequence diagram is shown in Figure 26 below and Figure 27 below.

8.9.1 Initial Radio Unit Monitor Sourced by the FNE

8.9.1.1 FNE Actions

When directed by a console the FNE shall comply with the channel access procedures described in 4.3.1 above and generate the RAD_MON_CMD message indicating the target SU and initiating SU addresses. The message shall also include a Transmit Multiplier value. The FNE may record that the SU has a pending Radio Unit Monitor.

Independently, or at the request of the console, the FNE shall retransmit the RAD_MON_CMD message until a maximum number of polite retries (see 4.3.1 above), NPRetry, or a valid response is received from the target SU.

If there is no response, the FNE and/or console may consider the RAD_MON_CMD to have failed and may indicate this to the requesting User.

If an ACK_RSP_FNE message is received from the target SU to the requesting SU, the FNE shall cancel the RAD_MON_CMD process and return to normal processing for this addressed SU. The FNE may indicate that the Radio Unit Monitor process was successful for the addressed SU.

8.9.1.2 Target Subscriber Unit Actions

An SU may ignore the RAD_MON_CMD if it does not support the feature. The rest of this section covers the case where the receiving SU supports the feature.

An SU receiving a RAD_MON_CMD message with a target address matching the UID of the SU shall adhere to the channel access rules of 4.2.3 above and return an ACK_RSP_FNE addressed to the dispatcher as the Target, and with itself as the source. If the Transmit Multiplier value in the RAD_MON_CMD is non-zero, and the SU is programmed with a non-zero transmit time value for radio unit monitor, then the SU shall commence with either a group or a unit-to-unit transmission on the channel for the time duration set by the product of the Transmit Multiplier in the RAD_MON_CMD and the transmit time value programmed in the SU. The SU determines what type of call to make based on its programming or configuration data. If a group transmission is started, the SU shall use its currently selected group. If a unit-to-unit transmission is started the SU shall use the source UID of the RAD_MON_CMD as the destination UID of the unit-to-unit transmission. After the time duration expires, the SU shall terminate the transmission.

8.9.1.3 FNE Response to ACK_RSP_FNE Sourced by an SU

The following actions are valid responses to an ACK_RSP_FNE, though a FNE need not perform all of these actions.

- (Repeat Mode Only) An FNE may transparently repeat the ACK_RSP_FNE as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.

8.9.2 Initial Radio Unit Monitor Sourced by a Mobile SU

8.9.2.1 Subscriber Unit Actions

The SU shall generate a RAD_MON_CMD to the FNE complying with the channel access protocol of 4.2.1 above. The RAD_MON_CMD shall indicate the SU address for the target SU and the initiating SU. The message shall also include a Transmit Multiplier value.

8.9.2.2 FNE Actions

The following actions are valid responses from the FNE to a RAD_MON_CMD though a FNE need not perform all of these actions:

- The FNE may transparently repeat the RAD_MON_CMD as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.

8.9.2.3 Initiating Mobile Subscriber Unit Actions

If a dispatcher mobile SU attempting radio unit monitor receives an appropriate response, then it shall take action as follows:

- If an ACK_RSP_FNE occurs that indicates acceptance of the service request by the target unit, the SU shall return to the idle state, and may indicate the success to the user.
- The SU shall retransmit the RAD_MON_CMD message until a maximum number of polite retries, NPRetry, or a valid response is received from the target. If it has exhausted its retries the originating SU shall return to the idle condition, and may indicate to the user the reason for the failure of the operation.

8.9.2.4 Target Subscriber Unit Actions

An SU may ignore the RAD_MON_CMD if it does not support the feature. The rest of this section covers the case where the receiving SU supports the feature.

An SU receiving a RAD_MON_CMD message with a target address matching the UID of the SU shall adhere to the channel access rules of 4.2.3 above and return an ACK_RSP_FNE addressed to the source as the Target, and with itself as the Source. If the Transmit Multiplier value in the RAD_MON_CMD is non-zero, and the SU is programmed with a non-zero transmit time value for radio unit monitor, then the SU shall commence with a unit-to-unit transmission on the channel for the time duration set by the product of the Transmit Multiplier in the RAD_MON_CMD and the transmit time value programmed in the SU. After the time duration expires, the SU shall terminate the call.

8.9.2.5 FNE Response to ACK_RSP_FNE Sourced by an SU

The following actions are valid responses to an ACK_RSP_FNE, though a FNE need not perform all of these actions.

- (Repeat Mode Only) An FNE may transparently repeat the ACK_RSP_FNE as defined in section 3 above. The FNE may also take console wireline priority into account when deciding whether to repeat the message.

8.10 Alert Tones

Alert Tone Procedures are the subject of future work.

9 Protection

Conventional LMR systems may support payload encryption and/or link layer encryption. Encryption of voice and data payload information is specified in [1]. Link layer encryption is the subject of future work.

Limited Use Only

10 Parameter Values

This section tabulates minimum, maximum, and default values for the parameters (where appropriate), e.g. time-outs or number of retries, used in this specification. A brief description of the usage of each parameter is given, but the precise definitions of usage should be obtained in this specification.

Limited Use Only

Parameter	Value: Max Default Min	Section	Description
FSSP	unspecified 728 ms 0	7.1	Frame Sync Seek Period. Time a receiver waits to acquire frame sync before it decides that the channel is idle.
NMax	20 5 1	None	Maximum number of supplementary data transmissions
NPRetry	15 5 1	8.1.2	Number of polite retries associated with a supplementary data transmission
NIRetry	5 0 0	8.1.2	Number of impolite retries associated with a supplementary data transmission
NTrafficLC	3 3 1	3	Rate at which the FNE shall repeat the source SU's embedded LC. Read as "every X LC" where X is the value of NTrafficLC
TAck	300 ms 200 ms 150 ms	4.2	Time that the SU waits before deciding it has not received the expected ack in time and should prepare to retry the message
TNextAttempt	300 ms 200 ms 150 ms	4.2	Time that the SU waits before starting its Holdoff timer and retrying after channel activity blocks it from sending a supplementary data message. Same as TAck except polite operation blocked it from sending a supplementary data message.
THoldoff	1000 ms (no default) 50 ms	4.2	Random amount of time an SU waits before transmitting a supplementary data message (TSBK) after deciding that it needs to retry a supplementary data message. A typical formula is: $50\text{ms} \times (1 + M\%N)$ Where M is a random number and N is user configurable from 1 to 20. Default value of N is 4. The larger the value of N the larger the range of possible values.
TNoAct	unspecified 3 s 0 s	2.2	Time span that no decodable message is present on the traffic channel. FNE may allow different TNoAct settings based on call type.

Annex A Inbound Signaling Packet (ISP) Names (Informative)

ISP Alias Name	Full Name
ACK_RSP_FNE	Acknowledgement Response – FNE (this data unit sent inbound and outbound on conventional channels)
CALL_ALRT_REQ	Call Alert Request
CAN_SRV_REQ	Cancel Service Request
EMRG_ALRM_REQ	Emergency Alarm Request
EXT_FNCT_CMD	Extended Function Command (this data unit sent inbound and outbound on conventional channels)
MSG_UPDT_REQ	Message Update Request
RAD_MON_CMD	Radio Unit Monitor Command
STS_Q_REQ	Status Query Request
STS_Q_RSP	Status Query Response
STS_UPDT_REQ	Status Update Request
TELE_INT_ANS_RSP	Telephone Interconnect Answer Response
TELE_INT_DIAL_REQ	Telephone Interconnect Voice Service Request (Explicit dialing)
TELE_INT_PSTN_REQ	Telephone Interconnect Voice Service Request (Implicit dialing)

Annex B Outbound Signaling Packet (OSP) Names (Informative)

OSP Alias Name	Full Name
ACK_RSP_FNE	Acknowledgement Response – FNE (this data unit sent inbound and outbound on conventional channels)
CALL_ALERT	Call Alert
DENY_RSP	Deny Response
EXT_FNCT_CMD	Extended Function Command
MSG_UPDT	Message Update
QUE_RSP	Queue Response
RAD_MON_CMD	Radio Unit Monitor Command
STS_Q	Status Query
STS_Q_RSP	Status Query Response
STS_UPDT	Status Update
TELE_INT_ANS_REQ	Telephone Interconnect Answer Request

Annex C Call Sequence Diagrams (Informative)

The following diagrams depict typical call sequences for voice and supplementary data calls. All diagrams assume each end point fully supports all messages associated with the feature. The “*” indicates the message is for illustration only.

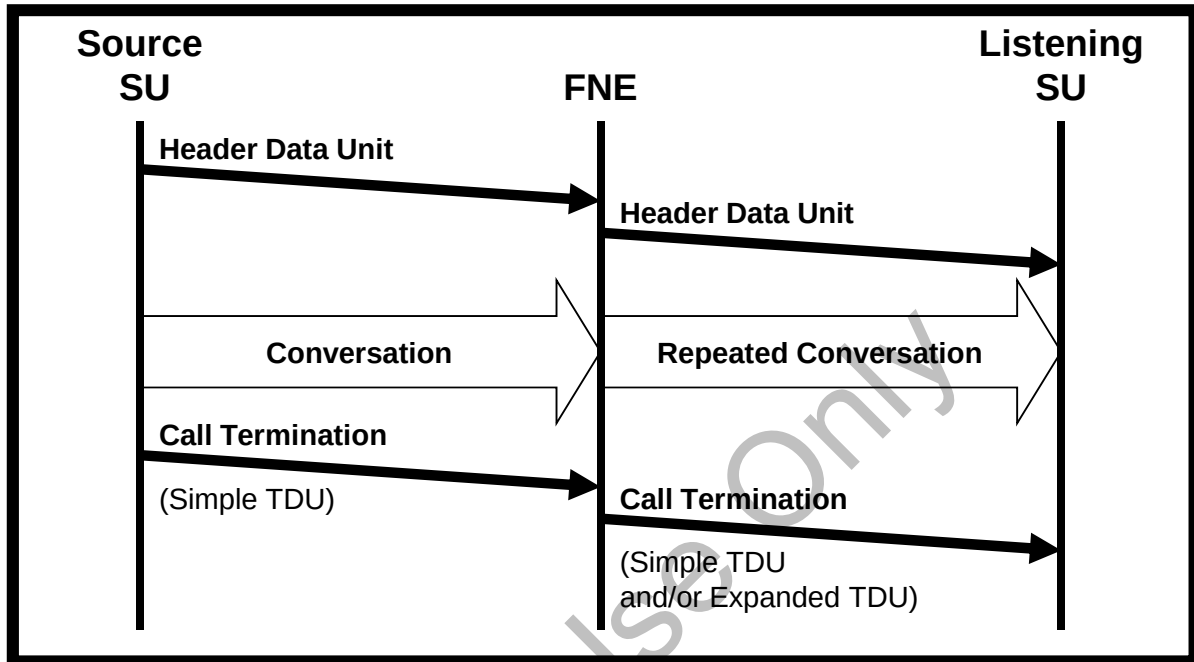


Figure 13– A Typical Call Sequence Diagram for Unit to Group Voice Call, Unit to Group Emergency Call, and Unit to Unit Voice Call

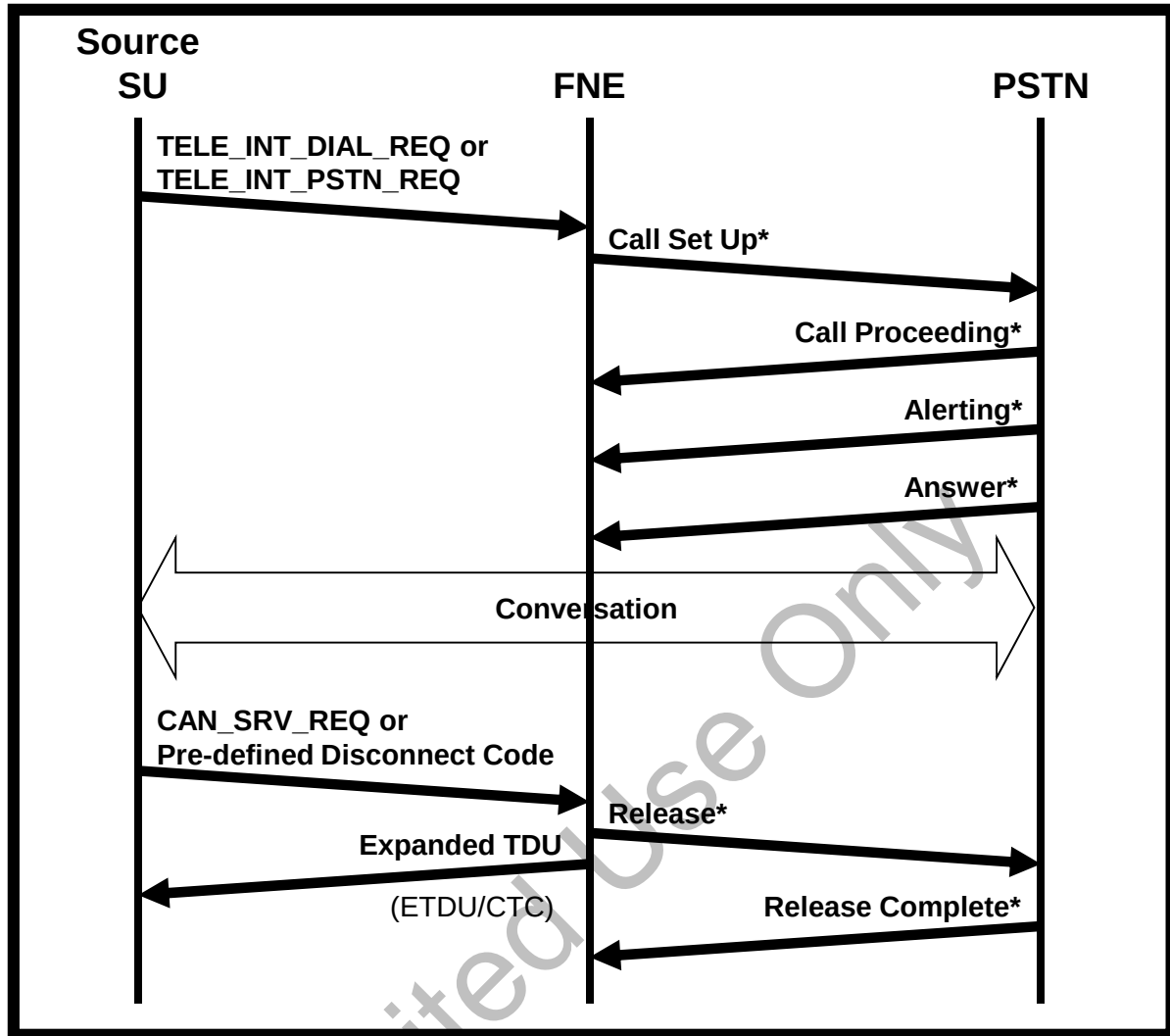


Figure 14– A Typical Call Sequence Diagram for Unit PSTN Voice Call

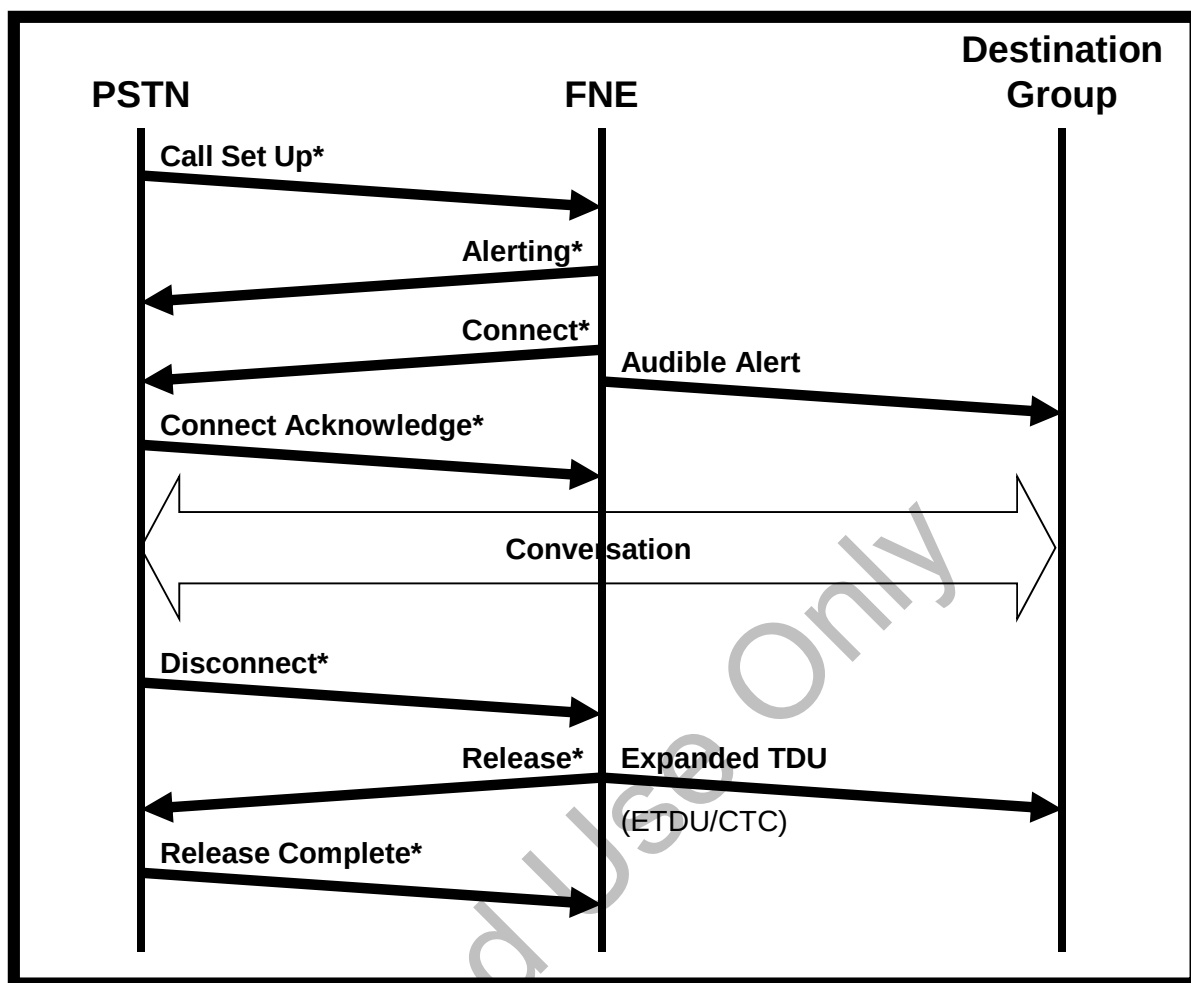


Figure 15– A Typical Call Sequence Diagram for PSTN to Group Voice Call

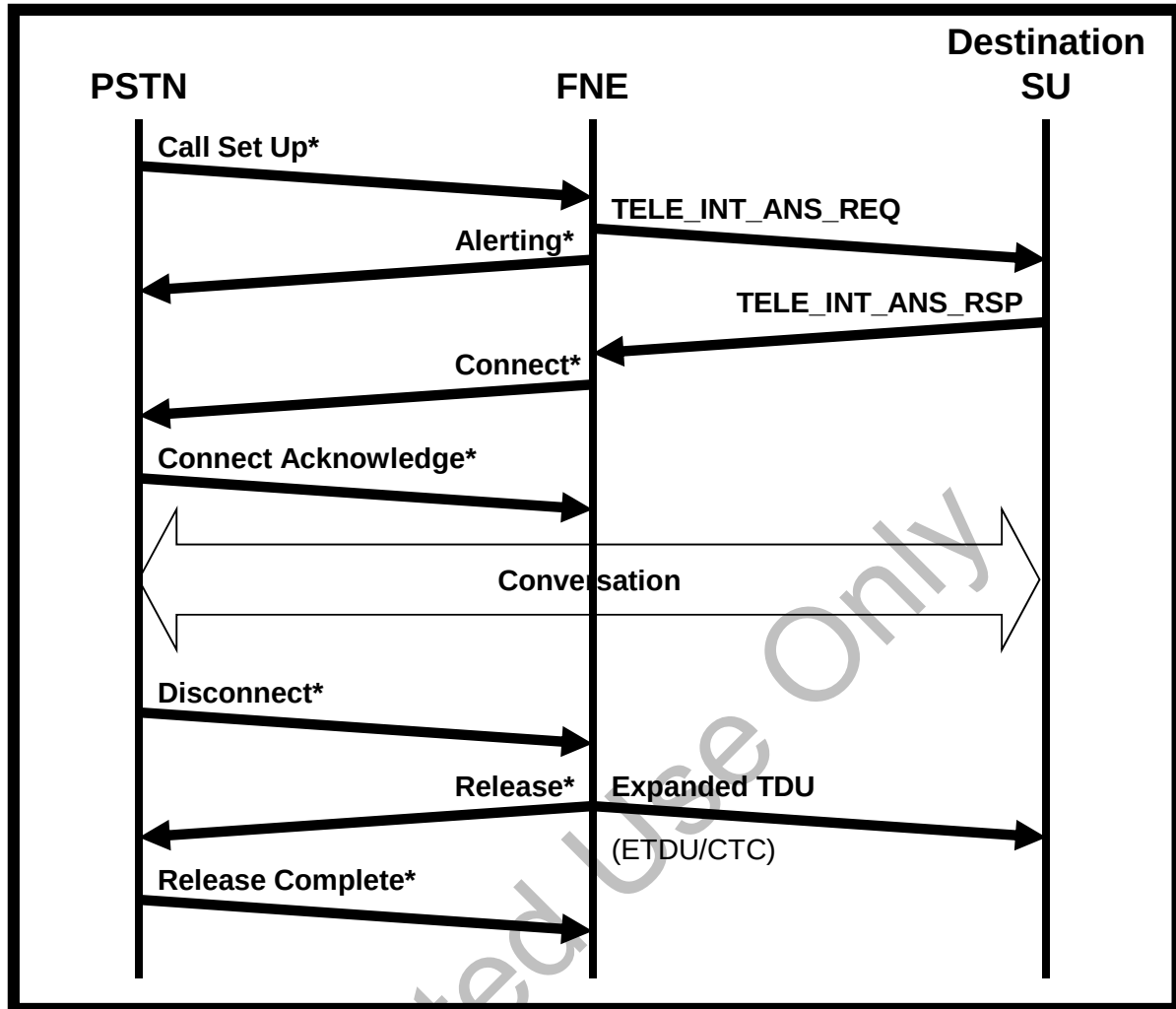


Figure 16– Typical Call Sequence Diagram for PSTN to Unit Voice Call

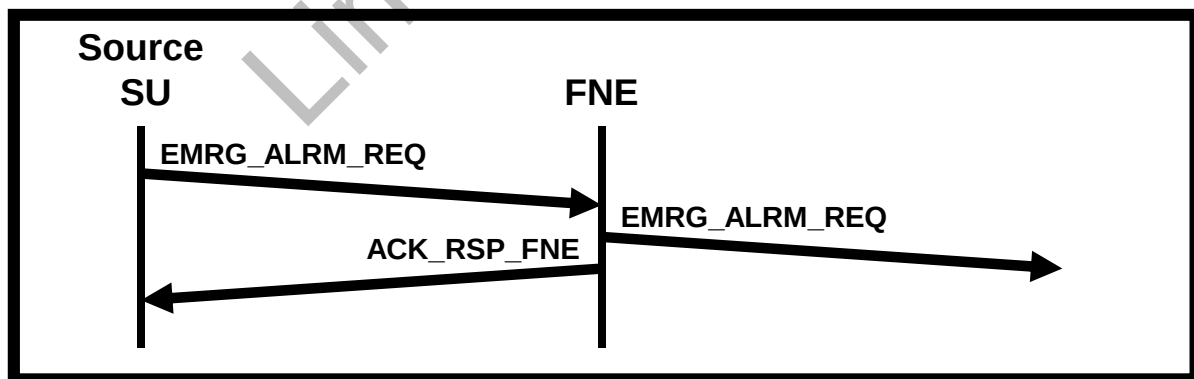


Figure 17– Typical Call Sequence Diagram for Emergency Alarm

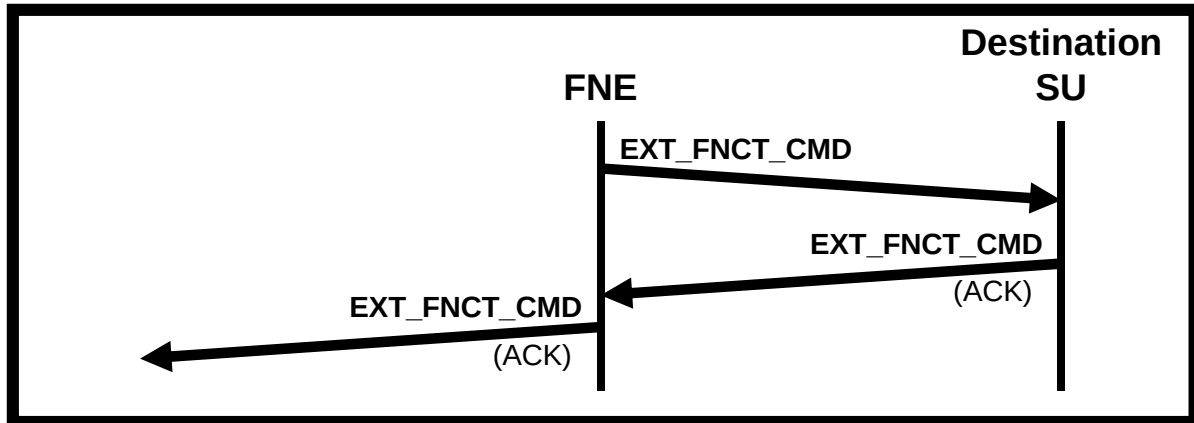


Figure 18– A Typical Call Sequence Diagram for Radio Inhibit (and Radio Check and Radio Uninhibit)

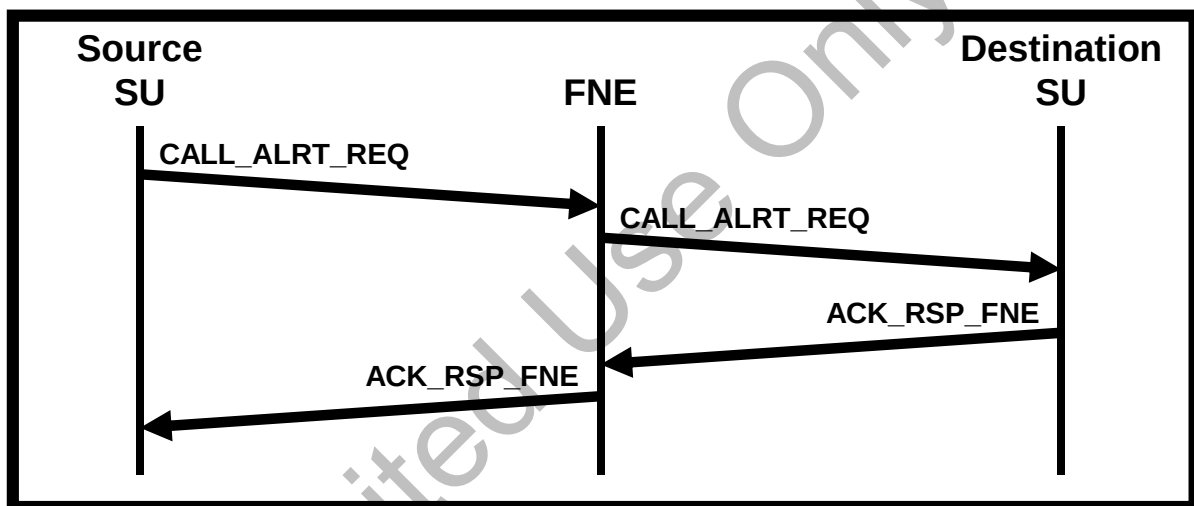


Figure 19– Typical Call Sequence Diagram for Call Alert Sourced by an SU

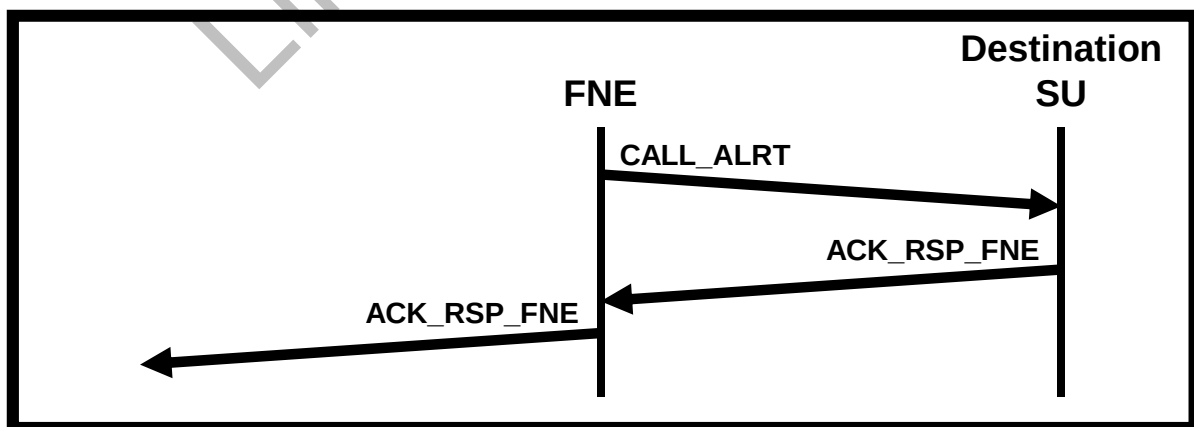


Figure 20– Typical Call Sequence Diagram for Call Alert Sourced by the FNE

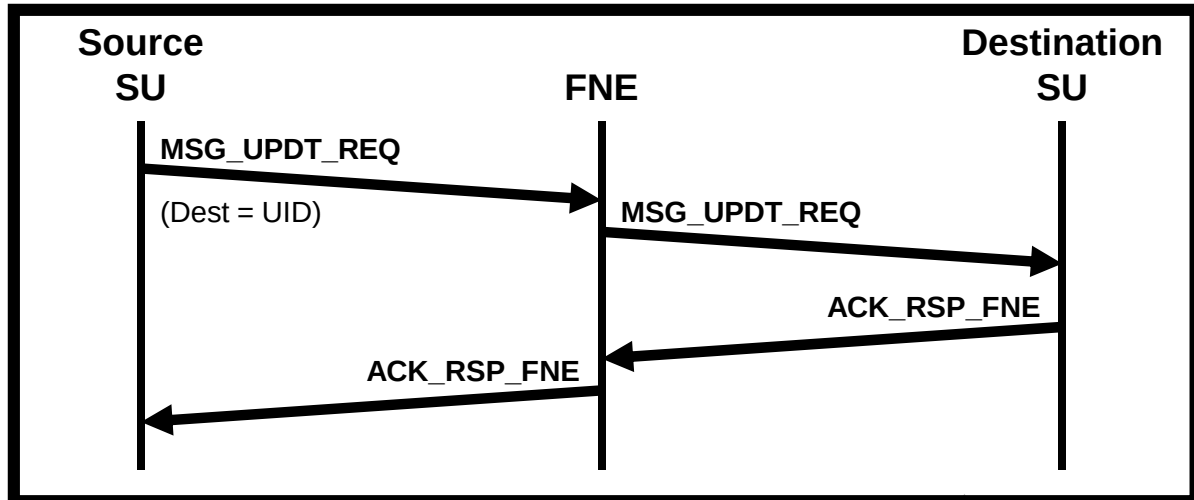


Figure 21– Typical Call Sequence Diagram for Message Update Sourced by an SU and Directed to a UID (Status Update is similar)

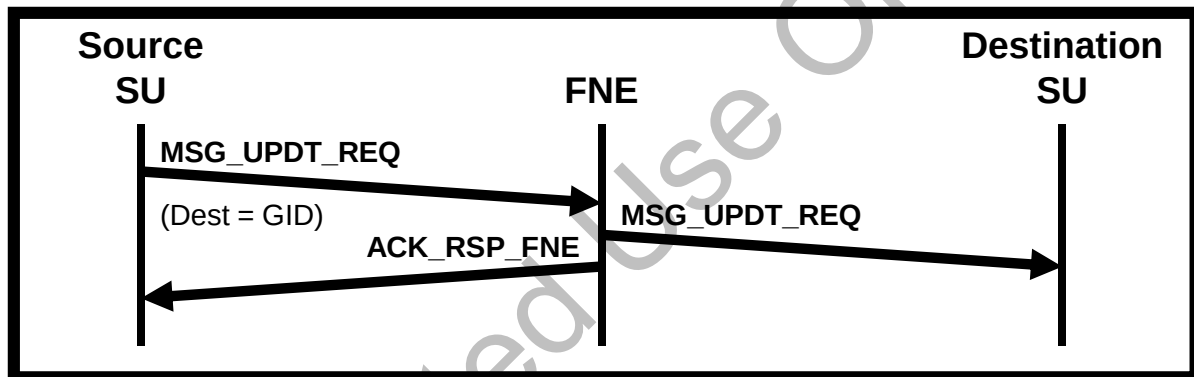


Figure 22– Typical Call Sequence Diagram for Message Update Sourced by an SU and Directed to a GID (Status Update is similar)

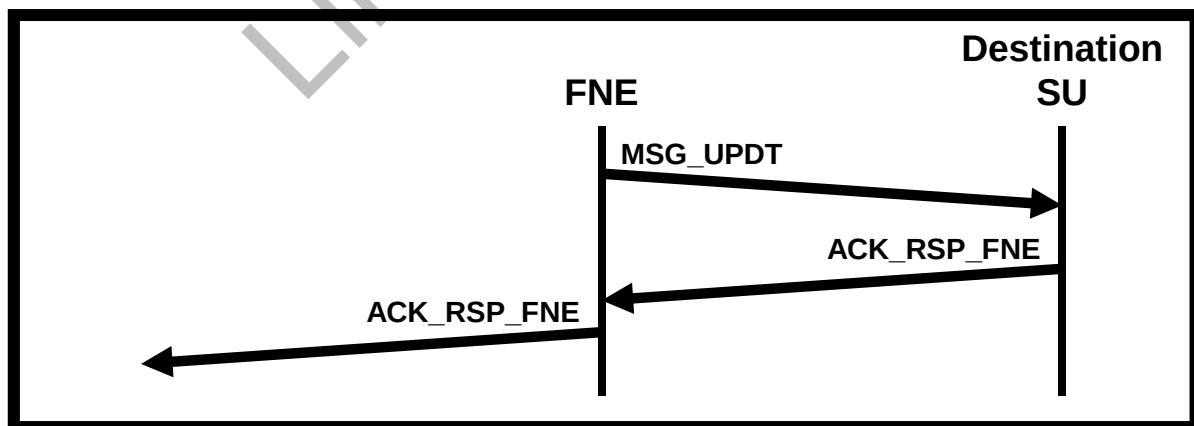


Figure 23– Typical Call Sequence Diagram for Message Update Sourced by the FNE (Status Update is similar)

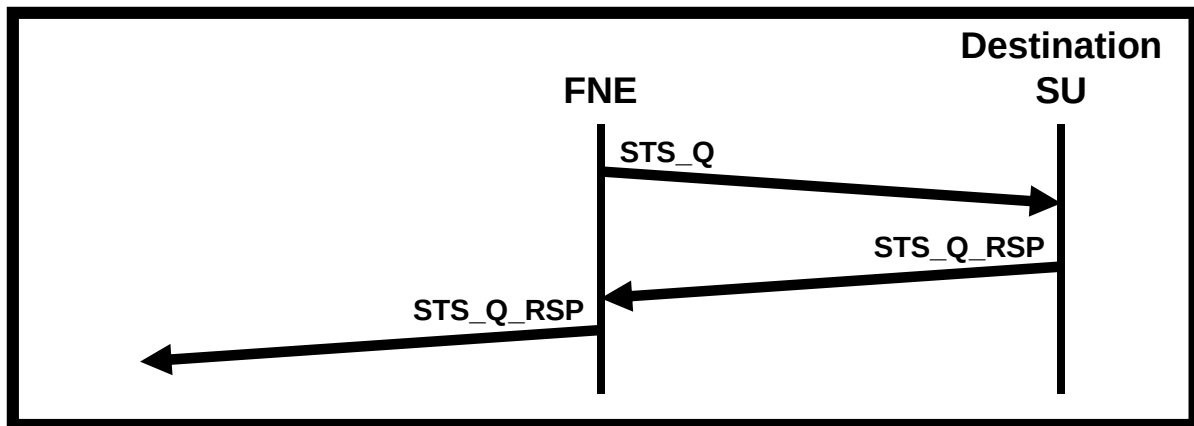


Figure 24– Typical Call Sequence Diagram for Status Query Sourced by the FNE

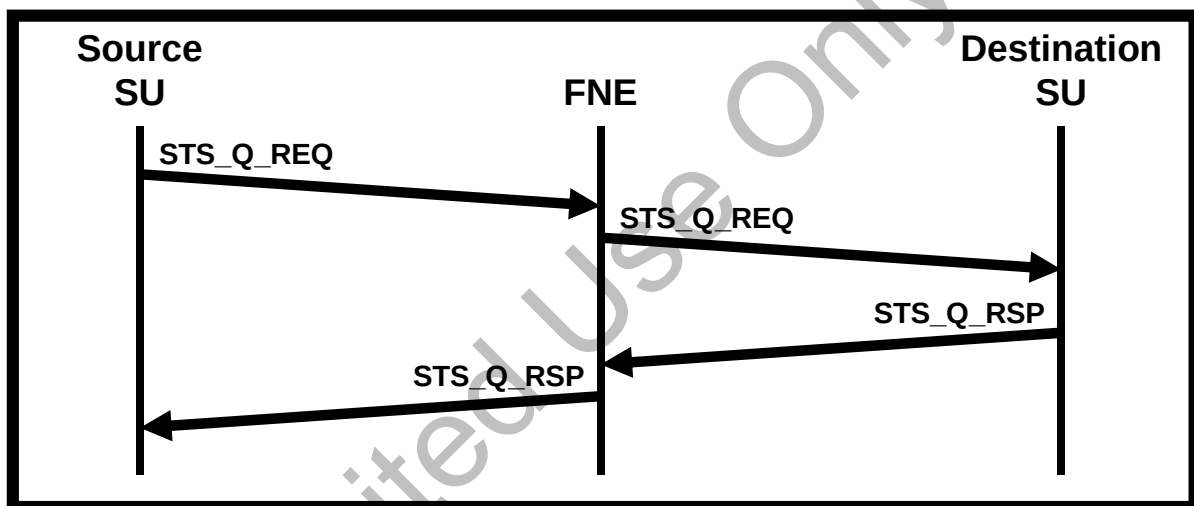


Figure 25– Typical Call Sequence Diagram for Status Query Sourced by an SU

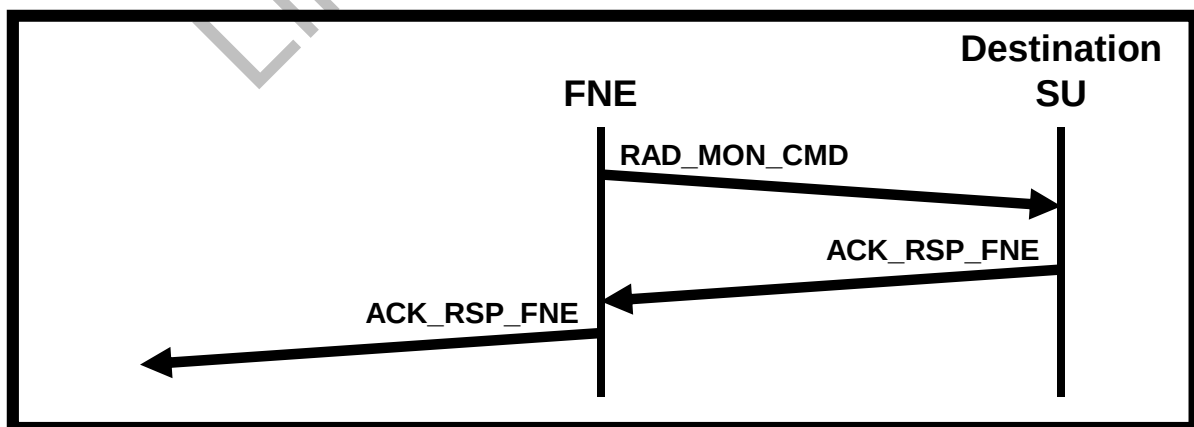


Figure 26– Typical Call Sequence Diagram for Radio Monitor Sourced by the FNE

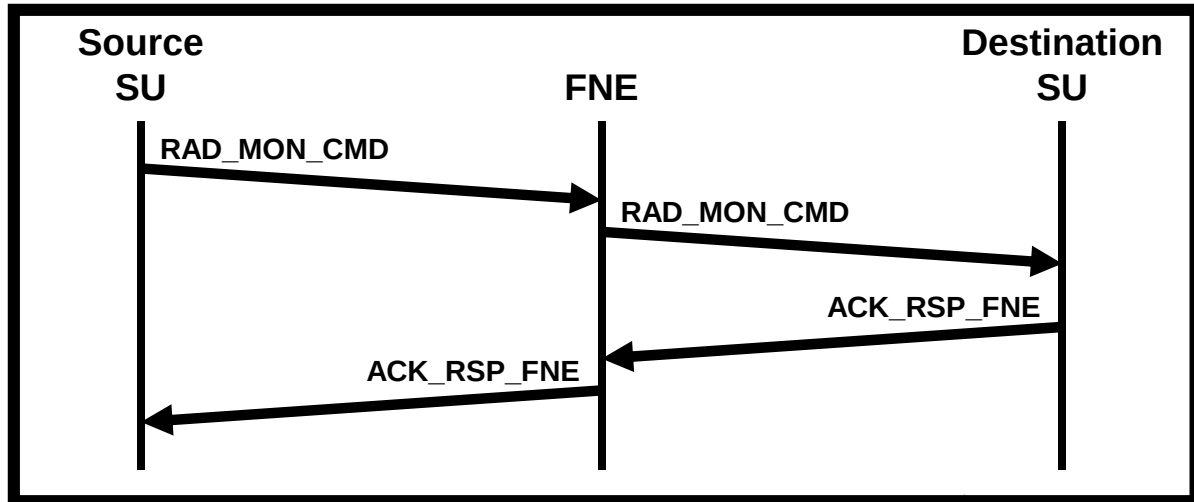


Figure 27– Typical Call Sequence Diagram for Radio Monitor Sourced by an SU

Annex D Transmission Preambles (Informative)

The SU can employ the use of preambles for voice and supplementary data transmissions.

The SU can transmit a preamble while it is ramping up its transmitter's power. If a preamble is employed, a recommendation is that it last no more than 30 ms and employ a high dotting pattern such as \$5F.

The SU can employ a preamble for impolite transmission of Emergency Alarm. A recommendation is that the preamble consists of a high dotting pattern (\$5F) and a length longer than an LDU (perhaps as long as 2 or more LDUs) so that if RF capture occurs the receiver has a chance to detect the frame synch of the new transmission.

Limited Use Only

Limited Use Only

Limited Use Only

THE TELECOMMUNICATIONS INDUSTRY ASSOCIATION

TIA represents the global information and communications technology (ICT) industry through standards development, advocacy, tradeshow, business opportunities, market intelligence and world-wide environmental regulatory analysis. Since 1924, TIA has been enhancing the business environment for broadband, wireless, information technology, cable, satellite, and unified communications.

TIA members' products and services empower communications in every industry and market, including healthcare, education, security, public safety, transportation, government, the utilities. TIA is accredited by the American National Standards Institute (ANSI).



**TELECOMMUNICATIONS
INDUSTRY ASSOCIATION**

**1320 N. Courthouse Road, Suite 200
Arlington VA, 22201, USA**

**+1.703.907.7700 MAIN
+1.703.907.7727 FAX**